Contract Pricing and Workflow Choice in Agentic AI

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Abstract

An agentic supplier chooses how many billable actions implement a customer's goal. We characterize when feasible bills permit profitable replacement of an inferior workflow. Class-specific action and goal prices implement the same optimal outcomes. A common action rate can make better steps lower user surplus, welfare, and completed goals. An execution fee changes adoption: in the adverse example, it lowers initial welfare and makes the subsequent quality comparison positive. General discovery conditions determine when supplier investment compensates conditional losses. The results identify how feasible pricing instruments shape the economic value of technical progress.

Keywords: agentic AI; contract pricing; workflow choice; action-based pricing; nonlinear pricing; technology adoption

1 Introduction

A customer buys a completed task, while an agentic service may bill intermediate actions. A software agent can resolve an issue through a compact intervention or a sequence of tool calls. A service agent can address a request directly or divide it into several operations. Their reliability and resource demands can differ. Customers may also face different checking requirements. When the supplier also chooses the implementation, the billing rule enters the production decision. The economic question is whether a more profitable workflow also creates more value for the customer.

Microsoft's Copilot Studio documentation lists action-related metering, and Salesforce's Agentforce pricing includes action-based credits alongside other charging arrangements (Microsoft, n.d.; Salesforce, n.d.). These arrangements illustrate the choice of billing units in agent services. The model studies a supplier that both receives the service bill and controls the offered workflow. An upstream technology platform may occupy a different position when its customer configures the agent and pays for the calls.

Technological decomposition gives the problem a concrete setting. ReAct interleaves reasoning and external actions, while Toolformer develops language-model use of external application interfaces (Schick et al., 2023; Yao et al., 2023). A sequence of calls can be productive even when the customer values only its completed result. SWE-bench evaluates the resolution of software issues, and WebArena evaluates functional completion of web tasks (Jimenez et al., 2024; S. Zhou et al., 2024). This distinction between action counts and endpoint value is central to the model.

This paper studies how feasible prices affect workflow choice and goal discovery. A monopoly supplier offers a complete workflow to each observable routing class before learning the customer's private value. Customers accept or refuse and choose how much to verify. The supplier pays the execution and preparation costs and invests in discovering serviceable goals. Billing capabilities determine which total prices it can attach to the offered workflows.

A bill-preserving replacement result establishes the contract benchmark. Suppose a coarse workflow gives customers weakly greater optimized value and costs weakly less than a fine workflow. Replacing the fine workflow at the same bill raises profit when the replacement is feasible and newly served customers have nonnegative margins. In the constant-cost model, positive fine-workflow profit supplies the margin condition. Discovery reoptimization preserves the ranking.

Class-specific prices remove the linkage between the two classes' bills. A class-specific action rate can reproduce any selected class-specific goal bill, giving the same optimal economic outcomes. The equivalence also holds with sequential workflow choice. In the coarse-dominance domain, a goal certificate supplies no additional profitable pricing capability when existing contracts remain available. Making the class label usable for billing can have positive private value.

A common action rate creates a different incentive. The low-value market constrains the rate, while an additional action raises the high-value market's bill. Better steps can then induce adoption of a longer workflow that remains technologically inferior. In a strict equilibrium construction, the unique optimal rate stays fixed and the supplier earns more by billing more actions. User surplus and correct goals fall, and extra resource use lowers welfare. The market-share condition identifies when low-value demand sustains that price restriction.

An execution fee changes when the supplier adopts the longer workflow. In both numerical economies, the fee-optimal supplier uses that workflow before the technical improvement, so subsequent improvement raises customer outcomes. In the adverse common-rate example, the fee lowers welfare at the initial technology and leaves welfare at the improved technology unchanged. The positive quality comparison therefore comes from an earlier adoption decision and a lower starting welfare level. This distinction between changing the contract and improving the technology is central to the welfare analysis.

Discovery links the execution decision to investment. Higher conditional profit finances more goal discovery, which can compensate a loss per discovery. With decreasing response elasticity, the return ratio determines whether compensation occurs at every cost, beyond one cutoff, or at no finite cost. Welfare has a separate boundary because discovery consumes resources. Exponential and hyperbolic discovery satisfy the condition under power costs; a convex-cost counterexample establishes the role of the elasticity restriction.

The analysis follows the bill through the supplier's implementation and discovery choices to customer value. Price flexibility affects the workflow from which technical progress begins, as well as the revenue available to finance discovery. Its private value and its social value can therefore differ.

1.1 Relation to the literature

The first connection is to quality choice and price discrimination. Mussa and Rosen (1978) study monopoly quality-price menus for privately heterogeneous consumers. Deneckere and McAfee (1996) show how quality degradation can serve price discrimination and can sometimes admit Pareto improvements. McAfee (2007) identifies conditions under which a damaged version is not profitable, including a proportional-value case under the maintained common-cost and regularity assumptions.

These contributions already explain why a supplier may manipulate quality and why welfare need not have one sign. The present analysis isolates a different constraint: the supplier can assign a workflow but may be unable to preserve its selected bill after replacing that workflow.

Versioning provides a precise comparison. Bhargava and Choudhary (2008) characterize profitable versioning for information goods with quality-invariant variable cost through the market shares of each version offered alone. Here customers receive one offered implementation within their routing class. Workflow assignment changes the feasible bill under a common action rate. The replacement result identifies when that restriction ceases to support an inferior implementation.

Nonlinear information-good pricing provides a closely related instrument comparison. Sundararajan (2004) analyzes usage pricing and an unlimited-use fixed-fee addition with private heterogeneity and usage-related transaction costs. The execution fee studied here is collected only when a discovered workflow executes. It changes the feasible pair of workflow bills without charging for entry into discovery. Ye and Zhang (2017) study how costly customer information acquisition affects monopoly nonlinear pricing. Our discovery decision belongs to the supplier, which receives the conditional return and bears the discovery cost. These timing and ownership differences determine which surplus is traded against discovery.

Complementary-service monetization also links an underlying benefit to intermediate revenue sources. White et al. (2024) study complementary websites whose advertising choices can distort volume and placement, with an additional payment instrument changing the outcome. That mechanism is an economic predecessor to the broader observation that one benefit may support several monetized activities. Here a single supplier controls the action count, and a bill-preserving implementation change determines whether an inferior workflow can remain profitable. Holmstrom and Milgrom (1991) similarly establish that incentives tied to measurable activities can distort task allocation and job design. The present allocation is the offered workflow, rather than an agent's hidden allocation of effort across tasks.

A second connection concerns diagnosis and contracting in expert services. The credence-goods literature separates knowledge of customer needs from the ability to verify provision and enforce contracts (Dulleck & Kerschbamer, 2006). Non-fraudulent provision can be sustained in appropriate

market environments (Emons, 1997), so an expert's informational advantage does not imply inefficient treatment by itself. Evidence on liability and verifiability also distinguishes these institutional features (Dulleck et al., 2011). Our type and goal certificates are deliberately separate. Their values follow from the bills each permits and from the supplier's retained alternatives.

The ownership of diagnostic information affects the incentive to produce it. Second opinions and price competition can weaken incentives for costly diagnosis (Pesendorfer & Wolinsky, 2003). Advice that customers reuse with discounters can induce experts to withhold it (Dulleck & Kerschbamer, 2009). More recent theory studies recommendations, prices, and consumer search jointly (Cao et al., 2026). In the present model the supplier owns the discovery decision and earns the conditional service return, which connects access to payment.

AI expert services extend these questions to delegated computational activity. Work on generative AI in credence markets examines how disclosure of agent objectives affects expert-market behavior (Erlei, 2025). PACT develops incentive-compatible and individually rational contracts for heterogeneous requirements and multidimensional agentic-service quality (Yang & Zhu, 2025). The present analysis examines which bills can be implemented through a workflow and how that constraint affects the return to discovery.

A third connection concerns delegation, verification, and complete decisions. Agrawal et al. (2018) distinguish prediction from the judgment needed to evaluate its consequences. Baird and Maruping (2021) develop a delegation framework for agentic information-system artifacts that can initiate activity and assume responsibilities. Proactive problem discovery is also an explicit technical objective in TIDE (Jeong et al., 2026). These contributions motivate an agent that can identify tasks and act on them. The model represents those capabilities through supplier discovery and execution.

Recent work places verification inside the economic response to AI. Bryan and Gans (2026) study training incentives in light of subsequent use and human verification. L. Huang et al. (2026) analyze delegation and verification under different objectives for the worker and the institution. Here the user chooses verification optimally and pays for execution. The supplier chooses the offered

implementation and finances discovery, creating a pricing incentive even when the user correctly evaluates the offer.

Accuracy and welfare can diverge for several established reasons. Algorithmic monoculture can reduce social welfare in a selection-market model even when the algorithms are more accurate (Kleinberg & Raghavan, 2021). Information congestion can make additional commercial messages unhelpful to their recipients (Anderson & de Palma, 2009). Workflow-dependent billing gives another channel: feasible prices govern implementation, and the resulting return governs discovery.

Section 2 defines the environment. Section 3 develops the contract benchmark, followed by the common-rate equilibrium and execution-fee comparison in Sections 4 and 5. Section 6 derives the discovery and welfare conditions. Section 7 studies verification, demand and costs, and Section 8 draws the economic implications. The appendices contain proofs and numerical comparisons.

2 Workflows, prices, and discovery

2.1 Timing and contractible events

A monopoly supplier serves a market of mass one. Potential customers belong to routing classes $z \in \{L, H\}$, with shares $\lambda_L = \lambda$ and $\lambda_H = 1 - \lambda$, where $0 < \lambda < 1$. Each customer has one potential genuine goal. The supplier observes the routing class before attempting to discover that goal. The class contains information about value but does not reveal the customer's eventual valuation. Conditional on discovery, value v has distribution F_z on a compact positive support.

The supplier posts a feasible price schedule before choosing discovery effort. For each class, it selects either no service, denoted OFF, or one complete workflow before the customer's value is realized. It then chooses discovery effort $s_z \geq 0$. A goal is discovered with probability $r(s_z)$, and the customer learns v. The customer can accept the offered workflow or refuse it. Upon acceptance, the customer chooses verification effort, the workflow executes, and its complete bill is paid. The supplier's workflow choice and discovery choice can equivalently be computed by backward induction at the posted schedule.

The supplier receives the customer's payment and chooses the service's internal workflow. The customer buys the offered implementation or exits; individual tool sequences are not separately purchasable. This is a delegated-service transaction in which implementation is part of the supplier's allocation. It separates the service provider's revenue from any upstream technology charges, which enter its execution costs.

The common action rate is a restriction on the feasible tariff. One interpretation is a service agreement that permits a common charge per action while leaving workflow assignment to the supplier. The supplier knows the routing class, but class-dependent rates become feasible only with a type certificate. Certification therefore changes the permitted billing schedule; the common-rate restriction is a maintained contractual rule. Research on disclosure credibility likewise distinguishes possession of information from its institutional use (Hu & Lei, 2023).

The two technological alternatives are coarse, C, with one action, and fine, F, with two. Write their action counts as $n_C = 1$ and $n_F = 2$. Four price schedules provide the initial comparison. Common action pricing, A, charges $n\tau$. Common goal pricing, G, charges one total p for either workflow. Class-specific action pricing, AS, charges $n\tau_z$. Class-specific goal pricing, D, charges p_z. All rates and total bills are nonnegative. Section 5 adds the execution-fee schedule B without changing the underlying technology or timing.

2.2 Verification and complete-goal value

A correct goal requires every action in the selected workflow to succeed. The customer has resource capacity B, incurs resource use c per action, and chooses variable verification $b \in [0, B - nc]$. Let $a > 0$ measure verification effectiveness, e_n the error probability before variable verification, and χ a common failure probability. Assume $B \geq 2c > 0$, $0 \leq \chi < 1$, and $0 \leq e_n \leq 0.35$. Conditional on execution, complete success has probability

$$P_n(b) = (1 - \chi)\left(1 - e_n \exp(-ab/n)\right)^n . \tag{1}$$

The common shock defeats the entire workflow. Conditional on its absence, the remaining action failures follow the multiplicative specification in (1). Verification is divided across the required

actions. An additional action therefore uses some capacity directly and spreads variable verification over more steps. This combination connects technical decomposition to the customer's total burden.

Workflow structure motivates the verification specification. Decomposed prompting and least-to-most prompting explicitly construct solutions from subproblems (Khot et al., 2023; D. Zhou et al., 2023). Software reliability research studies the selection of modular system designs under resource budgets (Berman & Ashrafi, 1993). The model combines that resource constraint with a common failure shock. Dependence across independently developed software is documented by Knight and Leveson (1986) and Littlewood and Miller (1989); recent language-model evidence also documents correlated errors across systems (Kim et al., 2025).

Let $\omega > 0$ denote the opportunity cost per unit of the customer's resource. Optimized value before payment is

$$h_n(v) = \max_{0 \le b \le B - nc} \{ vP_n(b) - \omega(nc + b) \}. \tag{2}$$

A total bill p is accepted when $h_n(v) > p$; refusal resolves exact indifference. Atomless values make that convention irrelevant for aggregate outcomes. The optimization in (2) depends on value and workflow, but an additive bill does not change verification conditional on execution. We write its optimizer as $b_n(v)$.

Verification here consumes a physical or organizational resource. Rational-inattention models instead allocate information-processing capacity under an information constraint (Sims, 2003). Attention management studies the allocation of costly attention across issues (Lipnowski et al., 2020). Those frameworks motivate a scarce user input, while equation (2) identifies the particular input used in this model: review that changes execution reliability. Technical work on learning to defer allocates cases between a prediction system and an expert (Mozannar & Sontag, 2020); the present customer instead chooses the intensity of checking within an offered workflow.

2.3 Conditional returns and discovery

Demand and surplus per discovered goal follow from the acceptance decision:

$$Q_{zn}(p) = \int \mathbf{1}\{h_n(v) > p\}\, dF_z(v), \qquad u_{zn}(p) = \int [h_n(v) - p]_+\, dF_z(v). \tag{3}$$

The expected number of correct goals per discovery is

$$x_{zn}(p) = \int P_n(b_n(v))\mathbf{1}\{h_n(v) > p\}\, dF_z(v). \tag{4}$$

These are conditional quantities. A customer with an undiscovered goal contributes zero to execution, payment, and realized benefit. Separating this discovery margin from success conditional on execution will be essential for the welfare comparison.

The supplier incurs real service cost $\kappa \geq 0$ per executed action. A fine workflow also requires preparation cost $\nu \geq 0$ per discovered goal, including discoveries followed by refusal. The coarse workflow has zero additional preparation cost. Its conditional profit is therefore

$$R_{zn}(p) = (p - n\kappa)Q_{zn}(p) - (n-1)\nu. \tag{5}$$

OFF has zero return and requires no preparation. Thus the supplier never needs to select a strictly negative-return active offer. Distinguishing costs incurred at discovery from costs incurred at execution prevents unsuccessful preparation from disappearing from the accounting.

Discovery costs $k_z\mathcal{C}(s_z)$ per potential customer in class z. Optimized profit per member of that class is

$$\Psi(R, k) = \max_{s \geq 0}\{Rr(s) - k\mathcal{C}(s)\}. \tag{6}$$

Both returns and discovery costs are multiplied by the class share when aggregated. The baseline has $r(s) = 1 - \exp(-s)$ and $\mathcal{C}(s) = s^2/2$. A positive return gives $s^* = W_0(R/k)$, where W_0 is the principal Lambert function (Corless et al., 1996). At nonpositive returns, OFF and zero discovery provide value zero. Section 6 replaces the baseline discovery functions while retaining the same conditional offers and accounting.

Discovery represents supplier effort to identify serviceable goals. It differs from a customer opening alternatives in a sequential search problem (Weitzman, 1979), and from customer entry or information acquisition under nonlinear pricing (Ye & Zhang, 2017). The distinction matters because the supplier's return determines discovery here. A price that transfers more surplus to the supplier can finance more goal identification, even when it reduces surplus per discovered goal.

2.4 Welfare and the numerical economies

For selected offers and optimized efforts, aggregate supplier profit and user surplus are

$$\Pi = \sum_z \lambda_z \Psi(R_z, k_z), \qquad U = \sum_z \lambda_z r(s_z^*) u_z. \tag{7}$$

Define welfare as $W = U + \Pi$, correct goals as $X = \sum_z \lambda_z r(s_z^*) x_z$, and billable actions as $N = \sum_z \lambda_z r(s_z^*) n_z Q_z$. Payments cancel in W. The remaining components are expected value from correct execution, less user resource use, service, preparation, discovery, and any purchased certification costs. These definitions value each correct goal by the customer's v, so a count of correct goals and welfare need not move together.

The numerical analysis uses normalized units and the parameters in Table 1. At the two error endpoints, verification uses all capacity for every value in the supports. Consequently, $h_n(v) = \alpha_n v - \omega B$, where $\alpha_n = P_n(B - nc)$. Appendix A derives the resulting demand. Along the full quality path, equation (2) also gives interior and zero variable verification.

The two economies share a model and differ in a small set of costs. This makes their different equilibrium responses informative about the same mechanism. All comparisons preserve the available coarse technology. Retaining that workflow ensures that an improvement in fine technology need not reduce the supplier's profit. A fixed-price technological ranking does not itself determine the optimized contract or the welfare ordering.

Table 1. Model primitives in the two numerical economies.

Primitive	Symbol	Example 1	Example 2
Low-class share	λ	0.8	0.8
Discovery costs	$k_L = k_H$	0.01	0.001
Service cost	κ	0.05	0.1
Preparation cost	ν	0.05	0.1
User resource cost	ω	0.02	0.01
Review capacity	B	1	1
Resource per action	c	0.1	0.1
Verification effectiveness	a	1	1
Common failure	χ	0.05	0.05
Coarse error	e_C	0.2	0.2
Fine error, before / after	e_F	0.35 / 0.10	0.35 / 0.10
Low value support		[0.35, 0.45]	[0.35, 0.45]
High value support		[1, 1.05]	[1, 1.05]

Notes: Values are uniform on the displayed supports. Baseline discovery is $1 - \exp(-s)$ with cost $ks^2/2$. Units are normalized per potential complete goal. The two economies differ only in the displayed primitives.

3 Price replication and contract capabilities

3.1 A feasible replacement at the same bill

The central question is whether the supplier can change implementation while preserving its bill. To isolate that question, consider general optimized customer values $h_{zC}(v)$ and $h_{zF}(v)$, execution costs $g_{zC}(v)$ and $g_{zF}(v)$, and preparation costs a_{zC} and a_{zF}. Let $E_{zn}(p)$ be the acceptance set at total price p. Conditional return is

$$R_{zn}(p) = \int_{E_{zn}(p)} [p - g_{zn}(v)]\, dF_z(v) - a_{zn}. \tag{8}$$

Suppose coarse implementation weakly raises optimized customer value and weakly reduces both costs in every state. Then $E_{zF}(p) \subseteq E_{zC}(p)$. At an unchanged bill, the exact return difference is

$$\begin{aligned} R_{zC}(p) - R_{zF}(p) = & \int_{E_{zF}(p)} (g_{zF} - g_{zC})\, dF_z \\ & + \int_{E_{zC}(p) \setminus E_{zF}(p)} (p - g_{zC})\, dF_z + a_{zF} - a_{zC}. \end{aligned} \tag{9}$$

The first term is a cost saving on existing customers. The second is the margin earned on newly accepting customers. The third is the preparation saving. This decomposition identifies why technological dominance alone is insufficient when additional demand can have a negative margin.

Call a contract set *closed under bill-preserving replacement* if replacing F by C keeps that class's total bill, all other classes' bills and allocations, and every remaining contractual restriction feasible. Closure includes any cross-class restrictions and incentive constraints.

Proposition 1 (Bill-preserving replacement). Suppose the contract set is closed under bill-preserving replacement, $h_{zC} \geq h_{zF}$, $g_{zC} \leq g_{zF}$, and $a_{zC} \leq a_{zF}$. If the coarse margin is nonnegative for almost every newly accepting customer, replacing F by C weakly raises conditional return. An increasing optimized discovery value preserves this profit ordering. With finitely many classes and an existing optimum, there is an optimum with no dominated active implementation. For a positive-mass class, suppose a term in (9) is strictly positive and optimized discovery value is strictly increasing over the relevant return interval. Then no optimum retains that active fine offer.

Equation (9) links technological dominance to the feasible contract set. At an unchanged bill, customer surplus weakly increases. Cost savings and nonnegative new-customer margins weakly raise the supplier's return. If optimal discovery is nondecreasing in return, aggregate user surplus and welfare also weakly increase along this same-bill deviation.

In the constant-cost model, the margin qualification is automatic for a positive-return fine offer. Since $R_F = (p - 2\kappa)Q_F - \nu > 0$, its execution margin is positive. Equation (9) becomes

$$R_C - R_F = (p - 2\kappa)(Q_C - Q_F) + \kappa Q_C + \nu \geq 0. \tag{10}$$

OFF handles nonpositive returns; zero return implies zero discovery. Appendix B gives the state-dependent-cost example that explains the new-customer margin condition and proves the strict replacement result.

3.2 Class prices and billing units

Class-specific action and goal prices produce the same selected bills. For any positive action count n_z, the maps

$$p_z = n_z \tau_z, \qquad \tau_z = p_z / n_z \tag{11}$$

are inverse on the selected workflow. Assume both contracts allow the same targeting, workflows, commitment, and discovery, and that their price labels have no separate cost or behavioral effect.

Proposition 2 (Class-price equivalence). Under these assumptions, AS and D have equal maximum supplier profit and the same sets of optimal on-path economic outcomes. The result also holds when the supplier chooses its workflow after posting the schedule. It does not require coarse technological dominance.

A globally optimal goal-price pair can be represented by its implied class-specific action rate. If a subsequent workflow change at that rate strictly raised profit, the resulting workflow and total bill would also be feasible under goal pricing and contradict global optimality. The reverse argument uses the second map in (11). Appendix B treats zero discovery and ties. The equivalence concerns optimal realized allocations, bills, and discovery; off-path workflow rankings can differ between the two schedules.

The distortion depends on which bills can change independently. A common action rate links the bills of different classes. Class-specific rates remove that linkage on selected workflows. Once that freedom is available, changing the bill's unit from an action to a goal supplies no further on-path pricing instrument in this environment.

3.3 The private value of contract capabilities

The two capabilities have different values in the dominance domain. Let a type certificate cost $K_T \geq 0$ and a goal certificate cost $K_G \geq 0$, both real fixed costs per market. Purchasing a capability retains all previously feasible offers. Common goal pricing has an all-coarse representative

optimum by Proposition 1. Common action pricing can reproduce its bill on the one-action workflow. Class-specific pricing includes common action pricing as a special case. Thus

$$\Pi_{AS} = \Pi_D \geq \Pi_A \geq \Pi_G. \tag{12}$$

Proposition 3 (Capability value). In the coarse-dominance domain, additive nonnegative certificate costs imply net optimal profit

$$\max\{\Pi_A, \Pi_D - K_T, \Pi_A - K_G, \Pi_D - K_T - K_G\} = \max\{\Pi_A, \Pi_D - K_T\}. \tag{13}$$

Type certification is purchased when $K_T < \Pi_D - \Pi_A$, is rejected when the inequality reverses, and admits both choices at equality. Goal certification has no strict incremental private value in this domain.

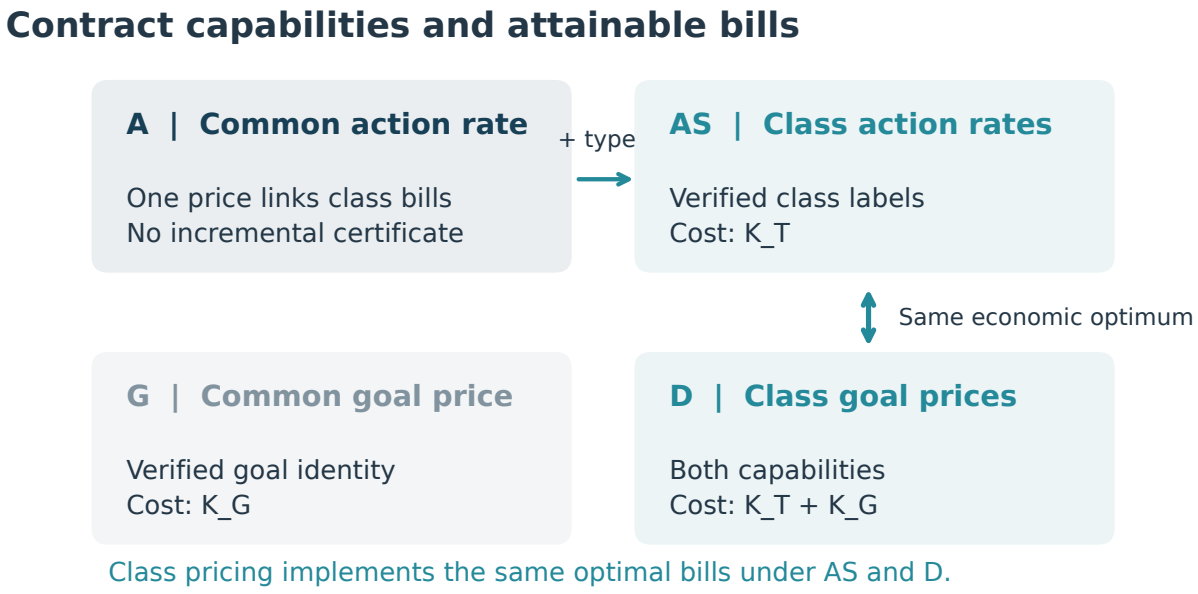


Figure 1. Contract capabilities and attainable bills. A type certificate permits class-dependent action rates; a goal certificate permits goal-denominated bills. Class-specific action and goal prices have the same optimal economic outcomes under Proposition 2.

A zero-cost goal certificate admits indifference. Requiring common goal billing and making that schedule available are different policies because voluntary availability preserves the supplier's action-price option. The private willingness to pay for type certification is

$$K_T^{\text{private}} = \Pi_{AS} - \Pi_A, \tag{14}$$

need not equal its social value $W_{AS} - W_A$. More flexible extraction can alter user surplus and discovery simultaneously. The figures and tables report those outcomes separately.

Outside the coarse-dominance domain, the last inequality in (12) can fail. A goal-only capability then has gross value $\max\{\Pi_A, \Pi_G\}$, so it may have private value before type certification. Proposition 2 still applies: conditional on feasible class-specific action pricing, an additional goal certificate remains redundant under its stated assumptions. This distinction will matter along the full technical-quality path, where the fine workflow can eventually become technologically superior.

4 Better steps under a common action rate

The restriction to a common action rate can make fragmentation profitable. A two-action workflow raises the customer's total bill without changing the posted rate. When low-value customers constrain that rate, the additional action can extract more revenue from high-value customers. A technical improvement can then induce adoption of a workflow that remains worse than the coarse alternative. This section establishes that mechanism with a globally optimized tariff and identifies the market structure that supports it.

4.1 A strict equilibrium construction

Consider the two numerical economies in Table 1. In each, the fine-step error falls from 0.35 to 0.10, while the coarse technology remains available. Both endpoints satisfy the coarse-dominance condition. At the optimized verification choices, success probabilities are

$$\alpha_C \approx 0.8727517646, \qquad \alpha_F^{\text{before}} \approx 0.5565278276, \qquad \alpha_F^{\text{after}} \approx 0.8269078164. \tag{15}$$

The improvement substantially raises fine-workflow reliability. Nevertheless, the improved two-action workflow still completes fewer goals per execution than the coarse workflow. The comparison therefore separates technical improvement within a workflow from the equilibrium choice between workflows.

Example 1 yields gains under a common action rate. Both before and after the improvement, the low-value class receives the coarse workflow and the high-value class receives the fine workflow. The optimal rate increases from 0.268264 to 0.285463. Aggregate user surplus rises by 0.033262, welfare by 0.052049, and correct goals by 0.054755. These favorable outcomes provide a useful comparison for the adverse construction, since they arise from the same timing and technology.

Example 2 produces a strict reversal. Its optimal tariff is

$$\tau^* = 0.35\alpha_C - 0.01 \approx 0.2954631176 \tag{16}$$

at both quality endpoints. Before the improvement, both classes receive the coarse workflow. Afterwards, the high-value class receives the fine workflow. Participation is complete almost surely in both equilibria. The low-value class is unchanged, so every aggregate difference comes from the high-value class's new implementation and discovery effort.

Proposition 4 (A strict adverse quality comparison). In Example 2, equation (16) is the unique global optimum under common action pricing at both quality endpoints. The improvement strictly raises supplier profit and billable actions, while strictly lowering user surplus, welfare, and correct goals. These properties hold on a nonempty relative open neighborhood of the example's continuous primitives, within the maintained timing, contract menu, and uniform-support family.

Appendix C proves global optimality by combining one-sided slope bounds near the candidate with profit bounds for more distant tariffs. Its strict comparisons also establish the relative open neighborhood in Proposition 4.

The supplier gains because it can charge twice the common rate. For the high-value class, conditional profit rises from $g = \tau^* - 0.1$ to $2g - 0.1$. That class's conditional success probability falls from α_C to α_F^{after}. Its higher return also increases discovery, which partially offsets the decline in success per

Table 2. Quality improvement under the optimal common action rate.

Outcome	Example 1	Example 2
Supplier profit	+0.018788	+0.018769
User surplus	+0.033262	-0.066850
Welfare	+0.052049	-0.048081
Correct goals	+0.054755	-0.008081
Billable actions	+0.005939	+0.198185

Notes: Each entry is the after-minus-before difference between separately optimized equilibria. Example 1 uses C/F at both endpoints; Example 2 switches from C/C to C/F. Welfare is gross of certificate costs, which are absent in this comparison.

discovered goal. The offset is insufficient in Example 2. More actions are billed even though fewer correct goals are completed.

The welfare loss has a real-resource component as well as a revenue component. Payments transfer surplus from users to the supplier and cancel from aggregate welfare. Extra execution and preparation consume resources, and the inferior workflow lowers expected gross value. Consequently, the increase in profit does not contradict the welfare decline. Section 7 decomposes these effects and shows how changing real costs can alter the initial workflow choice.

Example 2 is the first triple-adverse combination in the 648-point parameter enumeration, with identifier 341. Appendix G gives the domain and its outcome counts.

4.2 Which markets sustain the common tariff?

A large low-value class can support the tariff at its own demand kink. To state this precisely, let $L(\tau)$ and $H(\tau)$ denote each class's optimized profit contribution per potential customer. Each function includes workflow choice and discovery. Suppose τ_0 uniquely maximizes L on a compact tariff set. Define

$$d(\tau) = L(\tau_0) - L(\tau) > 0, \qquad b(\tau) = H(\tau) - H(\tau_0), \qquad \Theta = \sup_{\tau \neq \tau_0} \frac{b(\tau)}{d(\tau)}. \tag{17}$$

The ratio measures the high-value market's largest profit gain per unit of low-value profit sacrificed. Direct comparison of aggregate profit gives the exact support condition

$$\tau_0 \in \arg\max_{\tau}\{\lambda L(\tau) + (1-\lambda)H(\tau)\} \quad \Longleftrightarrow \quad \frac{\lambda}{1-\lambda} \geq \Theta. \tag{18}$$

A strict inequality makes the optimum unique. At equality, another tariff is also optimal only if it attains the ratio supremum. A supremum approached as τ tends to τ_0 need not create a second optimum. Appendix D also covers a nonunique low-market optimum and the resulting zero-denominator cases.

Demand curvature determines whether a finite supporting share exists. At a strict kink maximum, low-market losses are at least first order in a small tariff deviation. Locally Lipschitz high-market gains then imply a bounded ratio. By contrast, a smooth interior low-market maximum has a second-order loss. If the high-market profit slope is nonzero there, one direction produces a first-order gain and $\Theta = +\infty$. No share below one pins the common price exactly at that smooth monopoly optimum. The economic pressure from the low market remains, but exact tariff constancy is a stronger property.

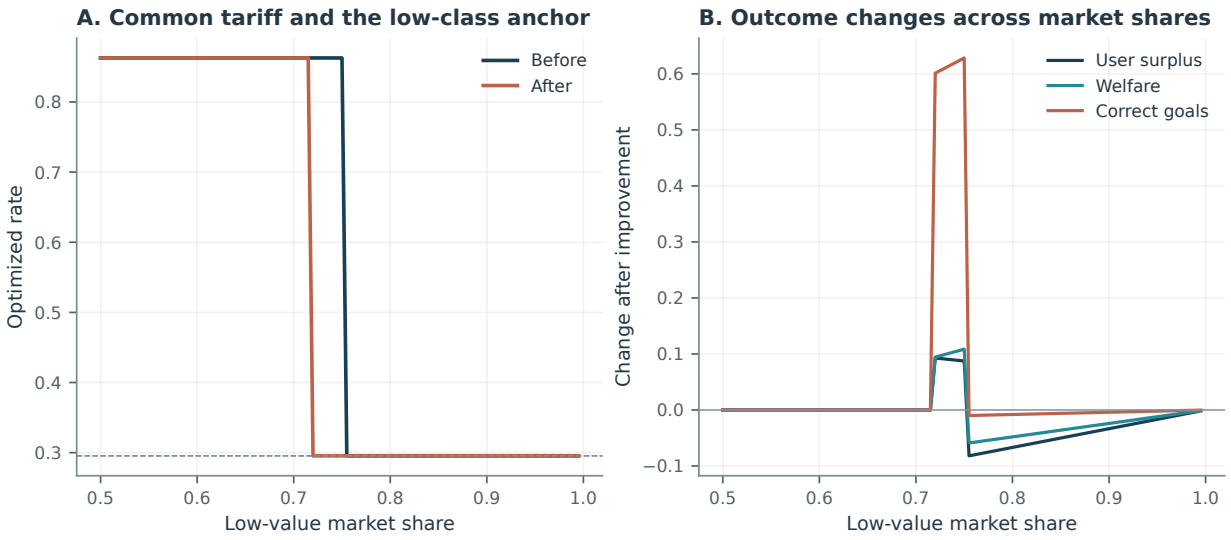


Figure 2. Market share and endogenous action pricing in Example 2. Panel A shows the optimal common rate before and after improvement; its dashed line is the low-market full-demand tariff. Panel B reports the outcome changes. Each share reoptimizes workflows and discovery. Displayed transitions are grid boundaries; equation (18) characterizes exact support.

The market-share path brackets the low-market tariff's onset between 0.750 and 0.755 before improvement, and between 0.715 and 0.720 afterwards. In the parameter grid, eight triple-adverse

comparisons occur at share 0.8 and 105 at share 0.95; none occurs at high-class discovery cost 0.1. A large low-value market constrains the common rate, while profitable discovery sustains service to the high-value class.

These results explain why tariff optimization does not automatically remove the distortion. The supplier trades off revenues across classes. A common rate ties their bills together, while workflow length offers a second way to change one class's charge. The next section tests whether a modest additional pricing instrument can replace that use of workflow length.

5 An execution fee and the feasible price set

An execution fee expands pricing possibilities without revealing customer class. Consider a common fee $f \geq 0$, paid only when a complete workflow executes, together with an action rate $\tau \geq 0$. Denote this contract by B. Its one-action and two-action bills satisfy

$$p_1 = f + \tau, \qquad p_2 = f + 2\tau, \qquad 0 \leq p_1 \leq p_2 \leq 2p_1. \tag{19}$$

Conversely, any pair in this cone has $\tau = p_2 - p_1$ and $f = 2p_1 - p_2$. Pure action pricing is the boundary $p_2 = 2p_1$; equal total bills form the boundary $p_2 = p_1$. The additional fee can reduce the ratio of the two bills while preserving their level. It therefore permits price discrimination through workflow assignment that pure proportional billing cannot reproduce.

The fee is collected after discovery and only upon accepted execution. Its payment date distinguishes it from the unlimited-use fixed fee in Sundararajan (2004). The customer continues to receive one supplier-selected workflow within its routing class.

5.1 Global solution of the fee problem

For a fixed mixed workflow profile, the objective is separable in total bills:

$$F_1(p_1) + F_2(p_2), \qquad F_i(p) = \lambda_i \Psi_i(R_i(p), k_i), \tag{20}$$

subject to the cone in (19). Under uniform values and binding verification, each positive-return component has one peak. Return is linear on full demand, concave quadratic on partial demand, and nonpositive beyond the support. The increasing discovery-value function preserves this ordering, with zero discovery available at nonpositive returns.

Proposition 5 (Execution-fee optimization). Suppose the two component objectives are continuous and unimodal, with finite maxima. A globally optimal mixed-profile bill pair can be found among: a feasible pair of independent component maximizers; the equal-bill boundary; and the pure-action boundary. Combining these candidates with all nine profiles in $\{\text{OFF}, C, F\}^2$ yields a global optimum of the full fee-and-workflow problem.

An interior constrained optimum can move toward a component's preferred bill without lowering profit. Such a move either reaches that preferred bill or hits a cone boundary. Repeating the argument for the other component proves the claim. Common-profile and single-active-class optima have boundary representatives. A strictly profitable continuation change in workflow would be another enumerated profile and contradict global optimality. Appendix E supplies the details and distinguishes representative optima from additional inactive-coordinate ties.

Proposition 5 reduces the fee problem to an independent-price candidate and two continuous scalar boundary problems, considered across the workflow profiles. The boundary prices are optimized over their full feasible intervals. Uniform demand supplies the component unimodality used in this reduction.

5.2 The two quality comparisons with an execution fee

The fee changes the comparison's starting point. In both examples, the optimal profile is already C/F before the technical improvement and remains C/F afterwards. Before improvement, the fee is 0.034398 in Example 1 and 0.044398 in Example 2, with a common action rate of 0.251065. Afterwards, the optimum lies on the pure-action boundary: the fee is zero. Its action rate is 0.285463 in Example 1 and 0.295463 in Example 2.

The fee changes the quality comparison by changing initial adoption. In Example 2, common action pricing gives welfare 0.333577 before improvement and 0.285496 afterwards. With an execution fee,

initial welfare is 0.230877, and final welfare is again 0.285496. The fee makes the longer workflow profitable earlier, lowering initial welfare by 0.102700. Subsequent technical improvement is then beneficial within that workflow. Table 3 separates these welfare levels from the quality changes.

Table 3. Welfare levels under action pricing and an execution fee.

Example	Contract	Before	After	ΔW
1	A	0.246992	0.299041	+0.052049
1	B	0.247216	0.299041	+0.051825
2	A	0.333577	0.285496	-0.048081
2	B	0.230877	0.285496	+0.054619

Notes: A is common action pricing; B adds an execution fee. Each entry is an equilibrium welfare level. The final column varies technology within a contract; comparing rows varies the contract at a fixed technology.

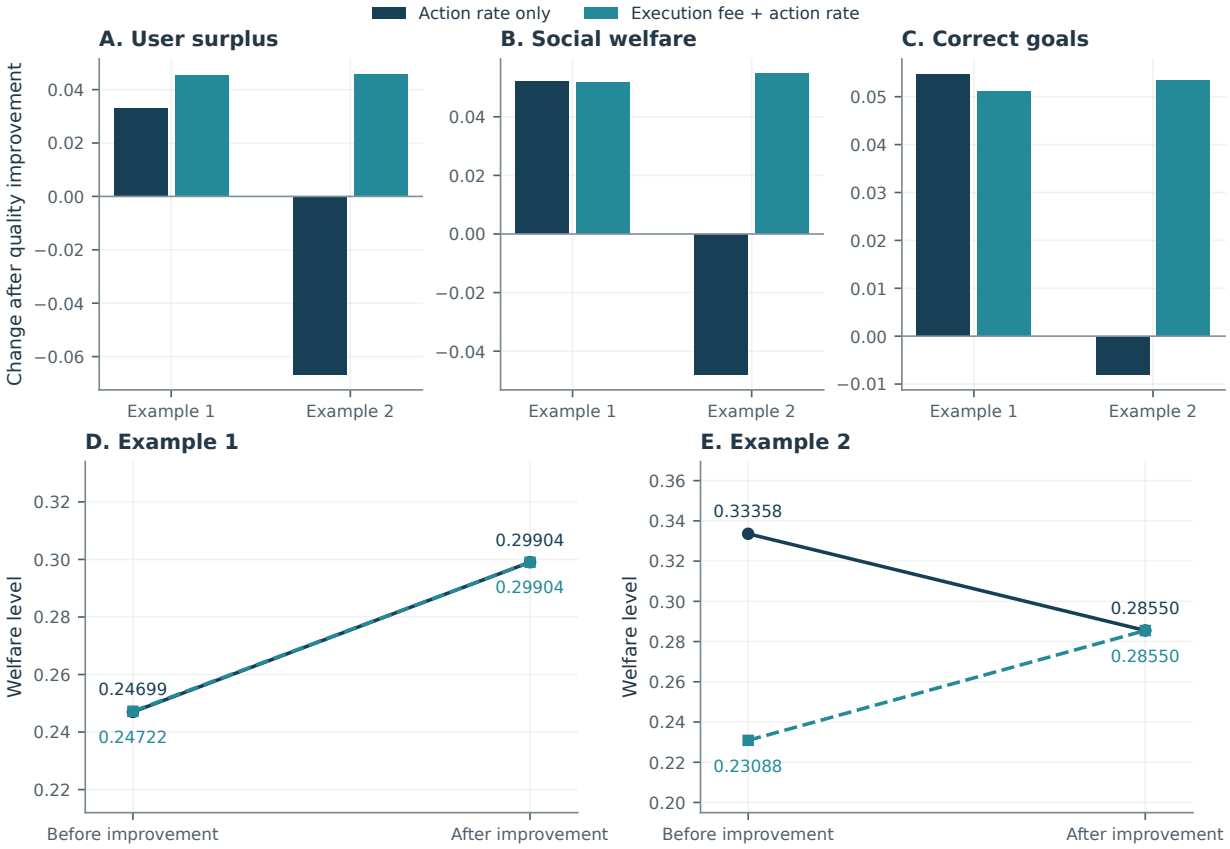


Figure 3. Execution fees and the starting equilibrium. The upper panels show quality-induced changes in user outcomes under each contract. The lower panels show welfare levels before and after improvement. In Example 2, the execution fee lowers initial welfare and leaves final welfare unchanged, reversing the sign of the quality comparison.

5.3 Endogenous type certification

The relevant outside option must include the execution fee when it is available. Since B contains pure action pricing, the private value of a type certificate becomes $\Pi_{AS} - \Pi_B$. In Example 2, this value falls from 0.1022153507 before improvement to 0.0934735723 afterwards. Write these cutoffs as K_+ and K_-, respectively. The certificate's fixed real cost is deducted from supplier profit and welfare once per market of mass one.

Three open cost regions exhaust the comparison. For $0 \leq K_T < K_-$, class-specific action pricing is chosen at both endpoints. Both classes use the unchanged coarse technology, so all outcome changes are zero. For $K_- < K_T < K_+$, the supplier uses class pricing before improvement and the execution-fee contract afterwards. User surplus rises by 0.0442784548, while net welfare changes by

$$\Delta W = K_T - (W_{AS}^{\text{before}} - W_B^{\text{after}}) \approx K_T - 0.0491951175 > 0. \tag{21}$$

Correct goals fall by 0.0104243382 in this middle region. At $K_T > K_+$, the execution-fee contract is used at both endpoints, and the three customer outcomes rise as shown in Figure 3. Both adjacent contracts are optimal at a cutoff; Table G6 reports their outcomes.

Across these cost regions, Example 2 has either unchanged outcomes or a gain in at least one customer outcome. The type-pricing equivalence determines the available contracts; their adoption thresholds follow from the returns of this economy.

The expanded price set separates the roles of the two observable events. Execution permits a fee, while a contractible class label permits different action rates. The private value of each capability is the gain over the best retained contract.

6 Discovery and welfare

6.1 How a return increase changes discovery

Discovery investment responds to the return on a service offer. To isolate that response, this section holds the two conditional offers fixed and varies discovery cost. Propositions 6 and 7 give the outcome and welfare cutoffs; the final subsection derives the additional response when prices adjust.

Retain equation (6) with $r(0) = \mathcal{C}(0) = 0$. Suppose the discovery optimum is unique and interior for each $t = R/k > 0$, and write $s(t)$ for the optimizer and $\rho(t) = r(s(t))$ for its discovery probability. At an optimum with a strictly negative objective second derivative,

$$tr'(s) = \mathcal{C}'(s), \qquad \varepsilon(t) = \frac{d\log\rho(t)}{d\log t} = \frac{t[r'(s)]^2}{r(s)[\mathcal{C}''(s) - tr''(s)]}. \tag{22}$$

The elasticity measures the response of discovery probability to a proportional return increase. Assume ρ is strictly increasing, ε is strictly decreasing, and

$$\rho(t) \sim a_0 t^{\eta} \quad (t \downarrow 0), \qquad \rho(t) \longrightarrow 1 \quad (t \uparrow \infty), \qquad a_0 > 0, \quad \eta > 0. \tag{23}$$

The first limit controls discovery when returns are small relative to cost. The second is saturation. Decreasing elasticity means that the same proportional return increase has less effect when discovery is already more profitable. Strict convexity of cost alone does not establish this property.

Compare fixed offers C and F with $R_F > R_C > 0$, and define $A_R = R_F/R_C$ and

$$H(k) = \frac{\rho(R_F/k)}{\rho(R_C/k)}. \tag{24}$$

Differentiating gives $d\log H/d\log k = \varepsilon(R_C/k) - \varepsilon(R_F/k) > 0$. Its limits are one as k tends to zero and A_R^{η} as k tends to infinity. Discovery is almost saturated in the low-cost limit, so the higher-return offer creates little additional coverage. At high costs, its relative coverage advantage approaches the power ratio determined by (23).

Proposition 6 (Discovery and a conditional outcome). Under (22)–(23) and strict decreasing elasticity, H is continuous and strictly increasing with range $(1, A_R^\eta)$. For positive conditional outcomes x_C, x_F, let $d = x_C/x_F$. The unconditional outcome under F is strictly lower than under C: never if $d \le 1$; exactly when $0 < k < k_x$ if $1 < d < A_R^\eta$, where $H(k_x) = d$; and for every finite $k > 0$ if $d \ge A_R^\eta$.

At a finite cutoff the outcomes are equal. If only x_F is zero there is always a loss; if only x_C is zero there is always a gain; if both vanish they are equal. Correct goals and nonnegative user surplus per discovery fit this classification at fixed conditional values.

6.2 A separate welfare boundary

Welfare includes the optimized cost of discovery. It therefore cannot be analyzed by treating welfare per discovery as a constant. Let $u_m \ge 0$ be user surplus per discovery under offer m, with $U_m = \rho(R_m/k)u_m$ and $W_m = U_m + \Psi(R_m, k)$. The envelope theorem yields

$$\frac{W_F - W_C}{\rho(R_C/k)} = u_F H(k) - u_C + \int_{R_C}^{R_F} \frac{\rho(R/k)}{\rho(R_C/k)}\, dR. \tag{25}$$

The integral is the supplier's optimized profit gain, expressed per discovery under the original offer. Each ratio in it strictly increases with k. Thus the normalized welfare difference is strictly increasing even if the new offer leaves zero customer surplus per discovery.

Define two conditional-value boundaries:

$$L_W = u_F + R_F - R_C, \qquad M_W = u_F A_R^\eta + \frac{R_F^{\eta+1} - R_C^{\eta+1}}{(\eta+1)R_C^\eta}. \tag{26}$$

Proposition 7 (Discovery and welfare). Under the assumptions of Proposition 6, $M_W > L_W$. There is no finite-cost welfare loss if $u_C \le L_W$. There is a unique finite cutoff k_W with loss exactly for $0 < k < k_W$ if $L_W < u_C < M_W$. There is a welfare loss at every finite $k > 0$ if $u_C \ge M_W$. Welfare loss implies user-surplus loss. If both cutoffs are finite, $k_W < k_U$.

The endpoint inequalities follow from the strict finite-cost range of the ratios. At $u_C = L_W$, the welfare difference is positive for every finite cost; at $u_C = M_W$, it is negative. Appendix F proves the limits in (25) using a uniform bound on the integrand. It also explains why no universal ordering follows between the welfare cutoff and a cutoff for correct-goal counts.

Discovery reoptimization has a positive welfare effect at the new offer. Let s_C and s_F be the respective optimal efforts, with $s_F > s_C$. Holding the new conditional offer fixed,

$$W_F(s_F) - W_F(s_C) = [\Pi_F(s_F) - \Pi_F(s_C)] + [r(s_F) - r(s_C)]u_F > 0. \tag{27}$$

The first term is positive because s_F uniquely maximizes the supplier's new profit. The second is nonnegative. Discovery therefore compensates part of the loss from an inferior workflow; Proposition 7 determines when that compensation is sufficient.

6.3 Primitive families and a counterexample

Two different discovery technologies satisfy the classification. With $\mathcal{C}(s) = s^p/p$, $p > 1$, exponential discovery gives

$$t = s^{p-1}e^s, \qquad \varepsilon(t) = \frac{s}{(s + p - 1)(e^s - 1)}. \tag{28}$$

Hyperbolic discovery, $r(s) = s/(1 + s)$, gives

$$t = s^{p-1}(1 + s)^2, \qquad \varepsilon(t) = \frac{1}{(p - 1) + (p + 1)s}. \tag{29}$$

Both have $\eta = 1/(p - 1)$ and strictly decreasing elasticity. Figure 4 plots the response ratios for $p \in \{1.5, 2, 3\}$ with conditional returns 0.25 and 0.45.

A convex-cost counterexample identifies the missing restriction. Keep exponential discovery and let

$$\mathcal{C}(s) = \frac{s^2}{2} + s + \frac{e^{-10s} - 1}{10}. \tag{30}$$

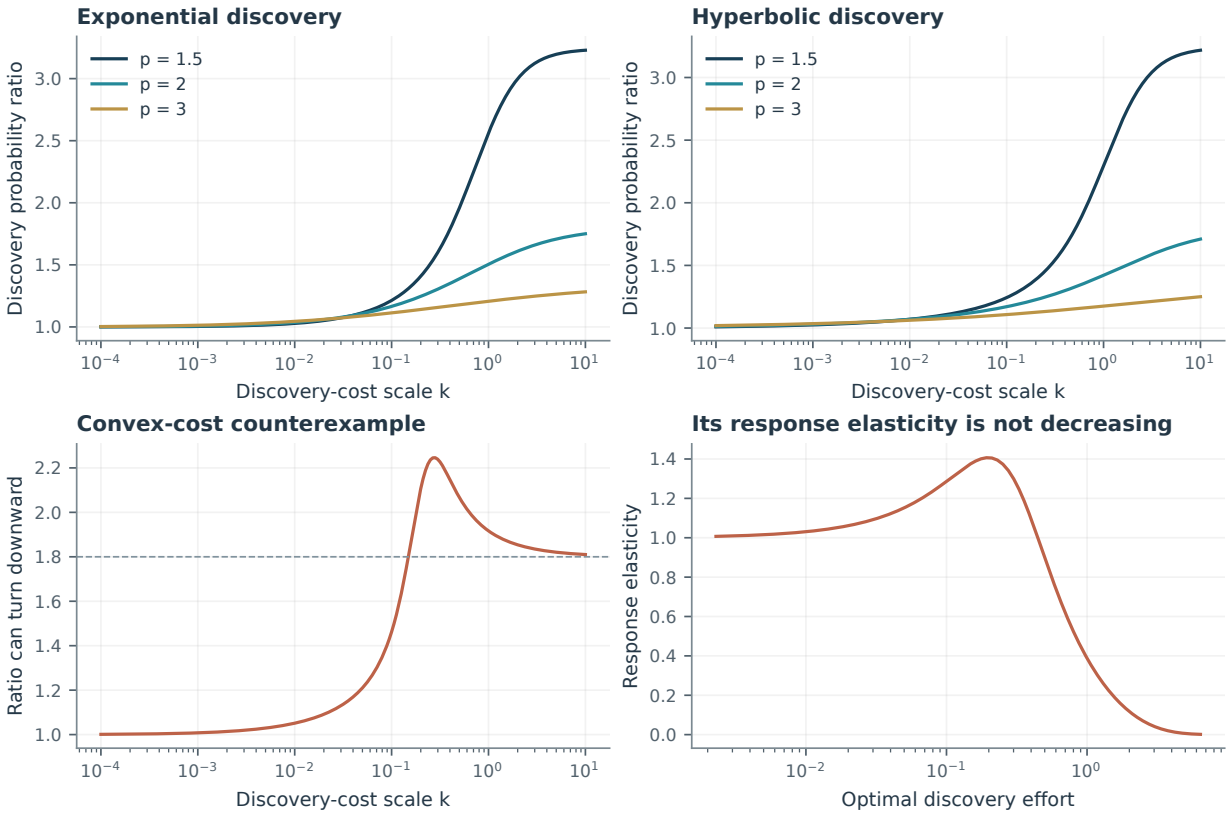


Figure 4. Discovery and the elasticity restriction. Exponential and hyperbolic discovery with power costs produce increasing response ratios. A separate strictly convex cost produces increasing elasticity near zero and a nonmonotone response ratio. The latter comparison shows why concavity and convexity alone do not justify the unique-cutoff conclusion.

Here $\mathcal{C}'(0) = 0$, $\mathcal{C}'(s) > 0$ for $s > 0$, and $\mathcal{C}''(s) = 1 + 10e^{-10s} > 0$. The discovery objective is strictly concave with a unique interior optimum. Nevertheless, $\varepsilon(t(s)) = 1 + 67s/22 + O(s^2)$ increases near zero. For large k,

$$H(k) = A_R \left[1 + \frac{67}{242} \frac{R_F - R_C}{k} + O(k^{-2})\right] > A_R. \tag{31}$$

The ratio eventually decreases toward its high-cost limit. This behavior isolates the economic role of decreasing response elasticity in the unique-cutoff result.

6.4 Endogenous prices introduce an additional response

The fixed-offer result has a precise boundary. On a smooth active tariff profile, suppose $R_z(\tau)$ is twice differentiable and independent of k_z at a fixed tariff. At a strict local optimum with $\Pi_{\tau\tau} < 0$, implicit differentiation gives

$$\frac{\partial \tau^*}{\partial k_z} = \frac{\lambda_z \rho(R_z/k_z)\varepsilon(R_z/k_z)R_z'(\tau^*)}{k_z \Pi_{\tau\tau}(\tau^*)}. \tag{32}$$

The tariff response has the opposite sign to that class's conditional-return slope. Higher discovery cost reduces the weight on its marginal revenue in the common-price decision. No common sign follows across classes or at a workflow switch.

If the compared returns themselves vary with a common cost k, the total derivative becomes

$$\frac{d\log H}{d\log k} = \varepsilon_C - \varepsilon_F + \varepsilon_F \frac{d\log R_F}{d\log k} - \varepsilon_C \frac{d\log R_C}{d\log k}. \tag{33}$$

Changes in conditional user surplus or success add their own derivatives. The equilibrium quality comparisons in Section 7 combine these responses with the optimized price and workflow. Equations (32)–(33) connect the fixed-offer discovery mechanism to that equilibrium adjustment.

7 Verification and demand along the technology path

7.1 Verification can become slack

Better steps reduce the marginal return to checking, so verification can become interior or zero. The optimal review decision in equation (2) connects the technology path to the customer's resource cost and acceptance decision.

The verification solution is explicit in the present error range. Let $x = e_n \exp(-ab/n)$ and $d = \omega/[v(1-\chi)a]$. For one action, an interior optimum has $x = d$. For two actions, the relevant root is

$$x = \frac{2d}{1 + \sqrt{1 - 4d}}, \tag{34}$$

when the marginal benefit at zero review exceeds its cost. The implied $b = (n/a)\log(e_n/x)$ is constrained to $[0, B - nc]$. If the marginal condition fails, variable review is zero; if the unconstrained choice exceeds capacity, review reaches its upper bound. At $e_n = 0$, variable review is zero. Appendix A proves these cases and the concavity that makes the solution global for the analyzed range.

Intrinsic self-correction can fail without additional feedback (J. Huang et al., 2024). Checkpoint models place verification within a fixed workflow and balance checking overhead against recovery opportunities (J. Zhou et al., 2026). Here the customer chooses total review, which changes the value of an offered workflow and its acceptance.

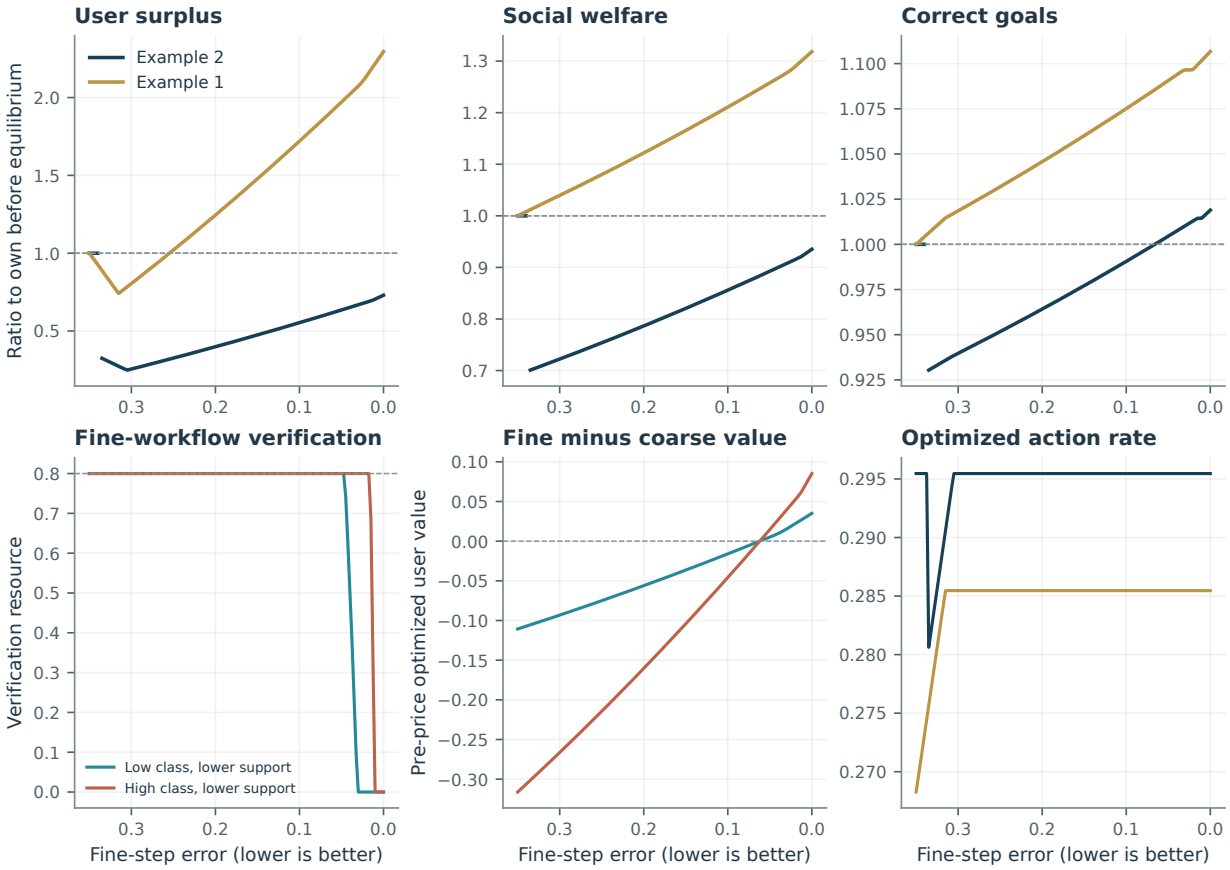


Figure 5. The full fine-step error path. The upper panels show ratios to each economy at error 0.35. The lower left and middle panels show Example 2 verification and fine-minus-coarse optimized value at each class's lower support. The lower right panel shows both optimal action rates. Every environment reoptimizes verification, pricing, workflows, and discovery.

The path uses 141 equally spaced errors from zero through 0.35 in each example. Example 2 changes from C/C to C/F between errors 0.3375 and 0.3350. Its correct-goal count recovers the initial level

at error approximately 0.0659122. User surplus and welfare remain below their initial levels even at zero fine-step error. Their endpoint changes are -0.0404066 and -0.0216377, while correct goals rise by 0.0161796.

Perfect fine steps therefore do not restore every outcome in this example. Zero idiosyncratic step error still leaves the common failure probability, and the chosen workflow still consumes service and customer resources. More fundamentally, endogenous payments and discovery can leave customers with less surplus even when complete success improves. Example 1 has the opposite endpoint pattern: its user surplus, welfare, and correct goals all rise at zero error. Its user-surplus path crosses the initial level at error approximately 0.2546223.

At sufficiently low fine-step error, the fine workflow can provide higher optimized value. The ranking $\Pi_A \geq \Pi_G$ then loses its coarse-dominance premise, while class-price equivalence $\Pi_{AS} = \Pi_D$ continues to follow from attainable bills.

7.2 Demand shape and exact tariff anchoring

Smooth demand changes the common-rate response. Both classes are assigned each of six Beta distributions, scaled to their unchanged supports: $(1, 1)$, $(2, 2)$, $(2, 1)$, $(1, 2)$, $(3, 1)$, and $(1, 3)$. Every shape is evaluated at both quality endpoints in both examples. The supplier reoptimizes its tariff, workflow assignment, and discovery for each environment.

Example 2 has triple-adverse changes under all six shapes. For Beta$(2, 2)$, its tariff increases from 0.3038985 to 0.3057064, while user surplus, welfare, and correct goals change by -0.0706502, -0.0500305, and -0.0162298. Thus the adverse direction can survive without an exactly unchanged tariff. This is consistent with the market-share argument: smooth demand can remove exact price anchoring while leaving the economic pressure that makes extra actions profitable.

Example 1 is favorable under five shapes but adverse under Beta$(3, 1)$, where user surplus falls by 0.0682235, welfare by 0.0276329, and correct goals by 0.0125722. The shape of demand changes the profitability of marginal customers and the selected workflow. Table G3 reports the comparisons.

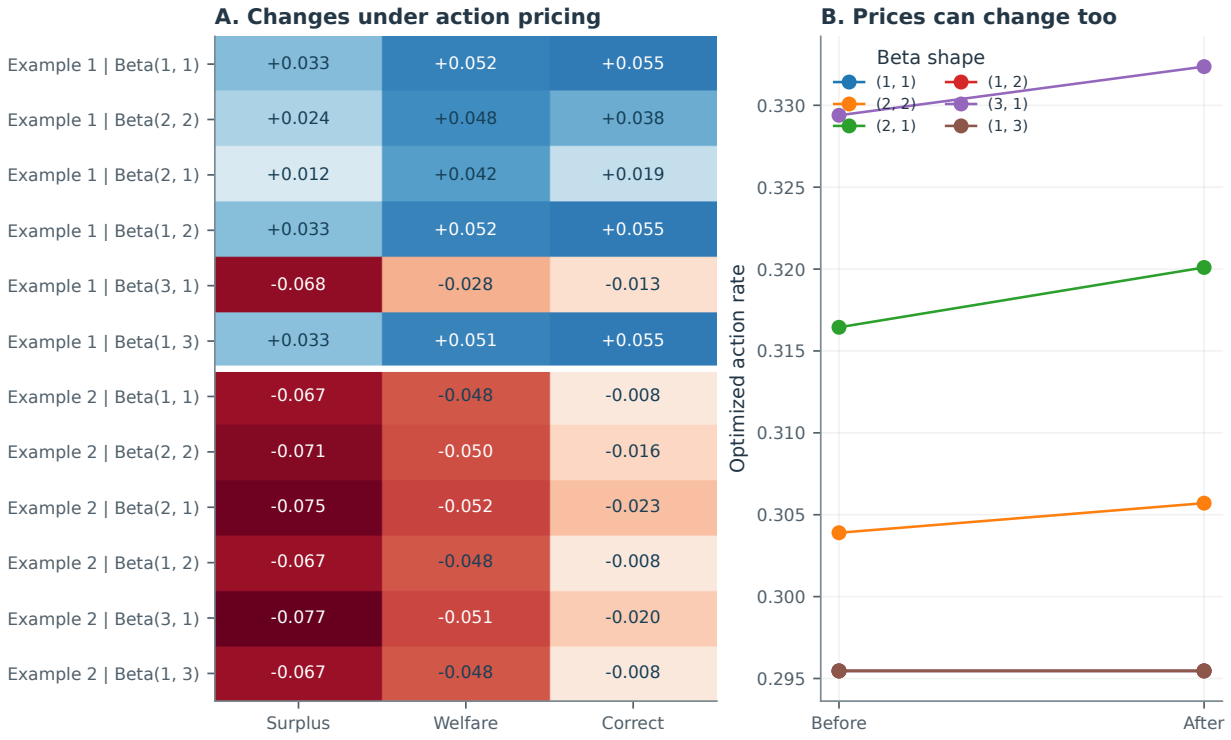


Figure 6. Demand shape and quality improvement. Panel A reports all six shapes for both examples, including the adverse Beta$(3, 1)$ comparison in Example 1. Panel B shows the optimal before and after rates for Example 2. Value supports are unchanged, and all endpoint equilibria are independently optimized.

7.3 Real costs and welfare accounting

The four cost comparisons distinguish service from preparation. Holding the remaining Example 2 primitives fixed, set each of κ and ν to either zero or 0.1. Both quality endpoints are reoptimized in all four cases. With both costs positive, the comparison is the C/C to C/F reversal. In each of the other cases, the profile is already C/F before improvement, and the quality gain raises welfare.

Table 4. Real costs and the quality comparison in Example 2.

(κ, ν)	Profiles	$\Delta\Pi$	ΔU	ΔW	ΔX
(0, 0)	C/F to C/F	+0.026301	+0.028716	+0.055017	+0.054350
(0, 0.1)	C/F to C/F	+0.026288	+0.028654	+0.054942	+0.054302
(0.1, 0)	C/F to C/F	+0.026160	+0.028721	+0.054881	+0.054993
(0.1, 0.1)	C/C to C/F	+0.018769	-0.066850	-0.048081	-0.008081

Notes: Every cost pair is solved at both quality endpoints. Profiles list low / high workflows. All remaining Example 2 primitives are held fixed.

In the adverse Example 2 comparison, execution and preparation contribute approximately -0.0198185 and -0.0197092 to the welfare change. Together they account for about 82 percent of

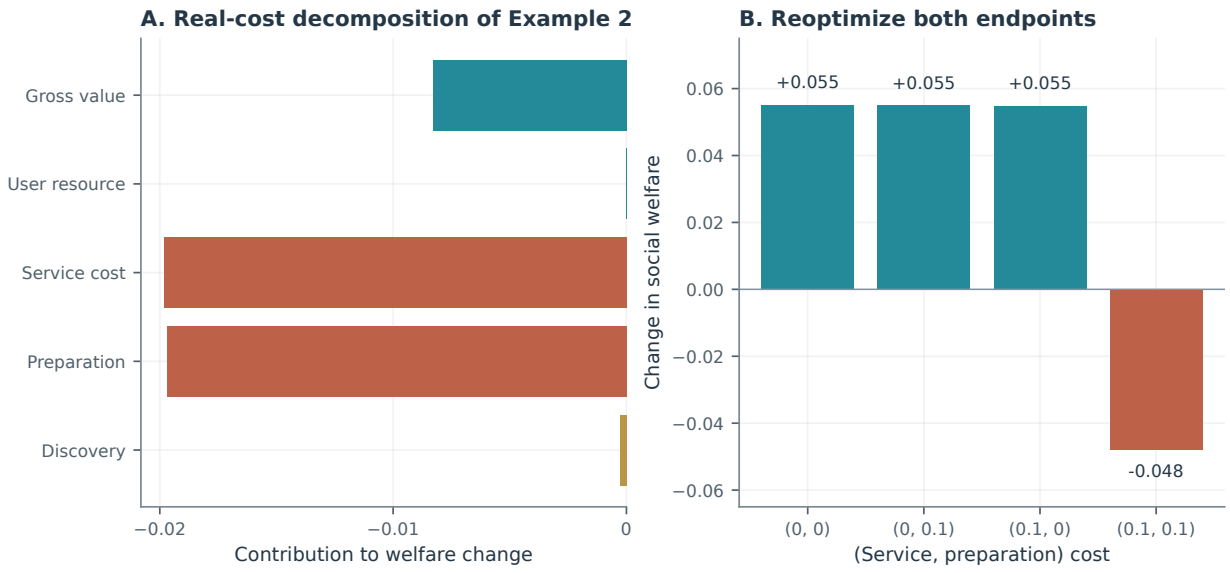


Figure 7. Real costs and welfare in Example 2. Panel A decomposes the welfare change into gross value and four resource-cost components. Panel B reports the complete welfare changes after service and preparation costs are independently removed, with both quality endpoints reoptimized. Payments cancel from welfare.

the 0.0480814 loss. Expected gross value contributes −0.0082828, customer resource use −0.0000109, and discovery −0.0002599. The components sum to the aggregate change within floating-point accuracy.

In these four cost comparisons, removing either cost changes the starting implementation. Elsewhere in the parameter grid, adverse outcomes also occur with one cost equal to zero. The relevant margin is the allocation supported by the joint technology and tariff, including the cost of serving each class.

Technical progress changes whole-workflow value through reliability and the cost of review. The supplier responds through its price, implementation and discovery choices. These responses explain why better individual actions can have different effects on complete goals and on customer surplus.

8 Economic implications and conclusion

8.1 Implications for service contracts

A service contract determines which bills can accompany a workflow. With a common action rate, an extra action raises the price of the complete service. A supplier that controls implementation

can use that link to collect more from high-value customers while preserving the rate that serves the low-value market. Bill-preserving replacement identifies the contract condition that removes this incentive.

Execution fees and class-dependent rates relax different pricing constraints. A fee changes the ratio of workflow bills, and class pricing separates the charges across customer groups. In the model, those rates implement the same optimal outcomes as class-specific goal prices. Certification has private value through the feasible bills it adds.

The welfare comparison depends on the starting allocation. Example 2's execution fee induces earlier adoption of the longer workflow, lowering welfare before technical improvement while leaving its improved-technology level unchanged. Type certification raises a further private-social wedge: after improvement its private value is 0.093474, while its gross social value is 0.049195. A cost between these values makes certification privately profitable and socially costly relative to the fee contract.

Discovery finances the extensive margin of service. When goals are already found with high probability, extra supplier return buys little additional coverage. With decreasing discovery elasticity, higher costs strengthen the relative coverage advantage of the higher-return offer. The user-outcome cutoff compares that advantage with the loss per discovery; the welfare cutoff also accounts for optimized discovery cost.

8.2 Scope of the analysis

The analysis applies to a supplier that controls the offered implementation under a specified tariff rule. Customers decide whether to purchase that service; they do not select from an unrestricted menu of implementations. The model also uses deterministic action counts and an execution-contingent bill. These contractual features delimit the replacement and pricing-equivalence results.

The technology has symmetric actions, with one common failure shock and conditionally independent remaining errors. Discovery identifies goals from the given value distribution, and the general cutoff result requires decreasing response elasticity. The numerical economies illustrate these mechanisms on the stated parameter domains.

Workflow choice links a billing rule to productive efficiency. A feasible same-bill improvement gives an undominated optimum, and strict profitable replacement excludes inferior active implementations. When bill constraints sustain a longer workflow, technical progress can trigger its adoption and lower customer value. The induced discovery response determines how far additional service coverage offsets that loss.

8.3 Code and computational materials

The accompanying replication supplement contains the model, numerical results and the code used for the tables and figures. Its README gives the reproduction command and the exhibit-to-output map.

Appendix A. Verification, participation, and accounting

A.1. Global verification choice

Fix an executed workflow and value v. With $x = e_n \exp(-ab/n)$, differentiation of (1) gives

$$P_n'(b) = (1-\chi)ax(1-x)^{n-1}. \tag{A1}$$

For $n = 1$, this marginal success gain decreases with b. For $n = 2$, its derivative is $-(1-\chi)a^2x(1-2x)/2$. Throughout the analyzed domain, $0 \le x \le e_n \le 0.35 < 1/2$. Thus the objective in (2) is concave, strictly so when the error and value are positive. An interior choice equates $vP_n'(b)$ to ω.

Let $d = \omega/[v(1-\chi)a]$. For one action, the unconstrained residual error is d. For two actions, an interior solution exists only when $d < e_n(1-e_n)$, which makes the zero-review derivative positive. The smaller quadratic root solves $x(1-x) = d$ and has the form (34), which avoids subtraction of nearly equal numbers. Otherwise set $b = 0$. In either case, constrain the implied review to the feasible interval. At zero error, positive review has no benefit and $b = 0$. These cases establish the full verification rule used in the computation.

The envelope theorem gives $h_n'(v) = P_n(b_n(v)) > 0$. Thus h_n is strictly increasing on the value support. A total bill either serves everyone almost surely, serves nobody, or has a unique marginal value v_p satisfying $h_n(v_p) = p$. Verification transitions occur when the marginal condition holds at $b = 0$ or at $b = B - nc$. Those transition values divide the support into segments on which integration and tariff bounds can be handled separately.

A.2. The binding-capacity endpoints

For the parameters in Table 1, coarse marginal benefit at maximum review is bounded below by

$$0.35(0.95)(0.2)e^{-0.9} > 0.02. \tag{A2}$$

For the fine workflow, maximum review is 0.8, and its least marginal benefit across the two error endpoints and all declared values is

$$0.35(0.95)(0.1e^{-0.4})(1-0.1e^{-0.4}) > 0.02. \tag{A3}$$

Concavity and these positive upper-boundary derivatives imply $b_n(v) = 1 - 0.1n$ for both endpoints and both examples. This conclusion is independent of tariffs and supplier costs, which do not enter the conditional verification problem. Every executed workflow then uses resource $B = 1$, and

$$\alpha_C = 0.95(1 - 0.2e^{-0.9}), \qquad \alpha_F(e) = 0.95(1 - ee^{-0.4})^2, \qquad h_n(v) = \alpha_n v - \omega. \tag{A4}$$

Coarse success exceeds fine success at both endpoint errors. Equality under this binding-capacity comparison occurs at

$$e = e^{0.4}\left[1 - \sqrt{1 - 0.2e^{-0.9}}\right]. \tag{A5}$$

Equation (A5) does not justify imposing binding review on the whole quality path. The full verification solution in A.1 is required when the fine error becomes sufficiently small.

A.3. Uniform demand and surplus

For $v \sim U[\ell, u]$, write $t = \omega + p$ and $D_n = \alpha_n(u - \ell)$. Direct integration yields

$$Q_n(p) = \begin{cases} 1, & t \le \alpha_n \ell, \\ (\alpha_n u - t)/D_n, & \alpha_n \ell < t < \alpha_n u, \\ 0, & t \ge \alpha_n u, \end{cases} \tag{A6}$$

and

$$u_n(p) = \begin{cases} \alpha_n(\ell + u)/2 - t, & t \le \alpha_n \ell, \\ (\alpha_n u - t)^2/(2D_n), & \alpha_n \ell < t < \alpha_n u, \\ 0, & t \ge \alpha_n u. \end{cases} \tag{A7}$$

Correct goals per discovery equal $x_n = \alpha_n Q_n$. Inserting demand into (5) gives the conditional profit. The supplier's real preparation cost remains present at zero demand if a fine offer is prepared, so OFF is evaluated as a separate option. Exact customer indifference affects a null set; supplier indifference between workflows can affect aggregate outcomes and is recorded separately.

Summing user surplus and conditional profit verifies transfer cancellation:

$$u_n + R_n = \mathbb{E}[\alpha_n v \mathbf{1}\{\text{execute}\}] - (\omega + n\kappa)Q_n - (n-1)\nu. \tag{A8}$$

Multiplying by discovery probability and subtracting discovery cost gives class welfare. With interior review, replace ωQ_n by expected $\omega(nc + b_n(v))$ over acceptors. This is the identity used for the five-component welfare decomposition.

Appendix B. Contract replacement and capability proofs

B.1. Proof of Proposition 1

Value dominance implies inclusion of acceptance sets. Split the coarse return integral into old and additional acceptors, subtract the fine return, and obtain (9). Cost dominance makes its first and third terms nonnegative. The margin assumption makes the second term nonnegative. Contract closure makes the comparison a feasible deviation. Applying the nondecreasing function Ψ preserves the conditional-return ordering, and repeating this operation over finitely many classes gives an undominated representative optimum.

Strictness follows if there is a positive-measure execution-cost saving among existing acceptors, a positive integrated margin among additional acceptors, or a strict preparation saving. Strict increase of Ψ over the return interval and positive class mass then make the deviation strictly profitable. If these conditions fail, a dominated implementation can remain tied. The weak theorem therefore concerns an available efficient representative, with efficiency defined by the stated technological dominance relation.

At a fixed bill, $[h_C(v) - p]_+ \geq [h_F(v) - p]_+$ pointwise. Conditional user surplus rises weakly. When discovery probability is nondecreasing in return, both the coverage and the nonnegative conditional

surplus weakly rise. Supplier profit also rises, so welfare rises. Independently reoptimizing bills is a different comparison and has no such user-surplus conclusion.

The margin qualification is necessary with state-dependent costs. Take two equally likely states, $p = 1$, and zero preparation cost. Fine implementation is accepted only in the first state; coarse implementation is accepted in both. Let fine costs be $(0, 3)$ and coarse costs be $(0, 2)$. Choose user values consistent with those acceptance sets and coarse value dominance. Fine return is $1/2$, whereas coarse return is zero. The new acceptor has a negative coarse margin. In the constant-cost application, $R_F > 0$ instead implies $p > 2\kappa$, which produces (10) and eliminates this counterexample. For $R_F \leq 0$, OFF supplies zero return and optimal zero discovery.

B.2. Proof of Proposition 2

For every selected workflow with positive action count, (11) is an invertible mapping between its total bill and its class-specific action rate. Hence the sets of feasible selected pairs (n_z, p_z) coincide. The two contracts also share conditional execution, discovery, and costs. Maximizing over identical selected-pair sets gives identical maximum profit and economic outcomes, including all maximizing pairs.

Sequential choice requires an additional continuation argument. Take a globally optimal goal-price pair (p, n) and post $\tau = p/n$. If the supplier could improve its payoff by choosing m at that action rate, the pair $(m, m\tau)$ would be feasible under goal pricing and strictly outperform its global optimum. Thus n is a best continuation response. Conversely, take a globally optimal action-price pair (τ, n) and set $p = n\tau$. A profitable goal-price continuation m at p could be represented by action rate p/m, which would contradict the action-price maximum.

The argument applies class by class. On positive returns, strict monotonicity of Ψ equates conditional-return and optimized-profit rankings. At nonpositive returns, select the sequentially optimal OFF or zero-return representative, with no discovery. Arbitrary labels for unused workflows do not change economic outcomes. Entire off-path strategies need not correspond, since alternative action counts change the bill under one posted schedule and leave it unchanged under the other.

B.3. Proof of Proposition 3

Class-specific action rates include a common rate, so $\Pi_{AS} \geq \Pi_A$. Proposition 2 gives $\Pi_{AS} = \Pi_D$. In the dominance domain, Proposition 1 supplies an all-coarse representative optimum under common goal pricing, after unprofitable offers are replaced by OFF. A common action rate equal to that goal bill reproduces the selected one-action offers. Allowing alternative continuation workflows cannot lower the optimized value. Therefore $\Pi_A \geq \Pi_G$.

With no capability purchased, the best retained contract has value Π_A. A type certificate supplies value $\Pi_D - K_T$. A goal-only certificate retains action pricing, so its best net value is $\Pi_A - K_G$. Purchasing both has value $\Pi_D - K_T - K_G$. Taking the maximum proves (13). At the type-cost threshold both alternatives remain optimal. A positive goal cost is strictly wasteful within this domain; zero goal cost creates possible acquisition indifference without positive incremental value.

If dominance fails, the goal-only value becomes $\max\{\Pi_A, \Pi_G\} - K_G$. The class-price mapping is unchanged, so adding goal certification after type pricing remains redundant. With an execution fee available before type certification, replace the initial gross value by Π_B. Because this contract contains the common-action and equal-total-bill boundaries, it retains the corresponding initial possibilities. Its numerical certification thresholds follow from the same net-profit comparison.

Appendix C. Proof of the strict action-pricing construction

The proof uses Example 2 and the uniform endpoint formulas. To shorten expressions, write $\alpha = \alpha_C$, $\beta = \alpha_F(0.35)$, and $\beta_+ = \alpha_F(0.10)$. Define

$$\tau_* = 0.35\alpha - 0.01, \quad g = \tau_* - 0.1, \quad D = 0.1\alpha, \quad T = 0.4375\alpha - 0.01. \tag{C1}$$

Both discovery-cost scales are 0.001, both real supplier costs are 0.1, and the low-class share is 0.8. All numerical inequalities below have strict rational enclosures in the executable certificate.

C.1. The low class uses the coarse workflow when active

Positive low-class fine demand requires

$$\tau < (0.45\beta_+ - 0.01)/2 < 0.2 = \kappa + \nu. \tag{C2}$$

At a common action rate, coarse demand weakly exceeds fine demand because its value is higher and its bill lower. For $\tau > 0.1$,

$$R_F - R_C = (\tau - 0.1)(2Q_F - Q_C) - 0.1 \leq \tau - 0.2 < 0. \tag{C3}$$

If $\tau \leq 0.1$, fine return is negative; it is also negative if fine demand is zero. Thus the low class's active positive-return offer is coarse. OFF handles the remaining prices.

C.2. All tariffs above the candidate are inferior

For $\tau \in [\tau_*, T]$, low-class demand is partial, with

$$R_L(\tau) = \frac{(\tau - 0.1)(0.45\alpha - 0.01 - \tau)}{D}. \tag{C4}$$

Its derivative is at most $1 - g/D < 0$. At the upper endpoint,

$$R_L(T) = (T - 0.1)/8 > 0.002. \tag{C5}$$

The optimal low-class discovery probability exceeds one half throughout the interval. Indeed $R_L/k_L > 2 > 2\log 2$, so the increasing function se^s implies $s_L > \log 2$.

For a high-class positive-return branch with n actions, differentiation gives $R' = nQ + (n\tau - n\kappa)Q' \leq n \leq 2$. Optimal discovery multiplies that derivative by a probability, so the high-class optimized contribution has derivative at most two. OFF contributes zero. Since the low-class derivative is negative and its discovery probability exceeds one half, the aggregate derivative obeys

$$\Pi'_A(\tau) \leq 0.8(0.5)(1 - g/D) + 0.2(2) = 0.8 - 0.4g/D < -0.09584. \tag{C6}$$

The objective is continuous and piecewise smooth with finitely many branch kinks. The almost-everywhere bound therefore makes it strictly decreasing over this entire interval.

For all $\tau \geq T$, low-class conditional profit is bounded by $R_L(T)$. The high class's conditional profit is bounded by its unconstrained coarse maximum $\alpha - 0.11$, using same-bill dominance and OFF. Since $\Psi(R, k) \leq R$ for nonnegative returns,

$$\Pi_A(\tau) \leq 0.8(T - 0.1)/8 + 0.2(\alpha - 0.11) < 0.179734. \tag{C7}$$

At τ_*, a feasible policy uses coarse workflows in both classes and effort four. It delivers

$$\Pi_A(\tau_*) \geq g(1 - e^{-4}) - 0.008 > 0.183883. \tag{C8}$$

Thus every tariff at or above T is strictly inferior at both quality endpoints. Together with (C6), this excludes every tariff above τ_*.

C.3. Tariffs below the candidate

After improvement, both high-class workflows have full demand for $\tau \leq \tau_*$ because $2\tau_* < \beta_+ - 0.01$. Their return envelope is $\max\{0, \tau - 0.1, 2\tau - 0.3\}$, which increases once positive. The low class has coarse return $\tau - 0.1$ when active. Every tariff below τ_* therefore produces strictly less aggregate profit than the candidate.

Before improvement, an additional local peak can occur at the high-class fine full-demand boundary,

$$\tau_f = (\beta - 0.01)/2. \tag{C9}$$

Below that boundary, both active high-class branches have increasing full-demand returns. Across its partial-demand interval, the fine-return derivative is bounded above by

$$2 - \frac{2(\beta - 0.21)}{0.05\beta} < -22.9. \tag{C10}$$

Whenever this fine branch is preferred, its return exceeds the low class's coarse return. The identical discovery costs then imply weakly higher high-class discovery. Consequently, the aggregate objective decreases on the preferred partial-demand fine branch. Once high-class coarse is preferred, both classes have return $\tau - 0.1$, and aggregate profit increases toward τ_*. A decreasing fine branch and increasing coarse branch cross at most once. It remains to compare τ_f with τ_*.

At τ_f, the active returns are $(\beta - 0.01)/2 - 0.1$ for the low class and $\beta - 0.31$ for the high class. Both exceed $0.003e^3$, so their optimal efforts exceed three. The weighted discovery cost exceeds 0.0045. Bounding discovered revenue by its full-discovery value gives

$$\begin{aligned} \Pi_A(\tau_f) &< 0.8[(\beta - 0.01)/2 - 0.1] + 0.2(\beta - 0.31) - 0.0045 \\ &= 0.6\beta - 0.1505 < 0.183417. \end{aligned} \tag{C11}$$

This is below (C8). We have excluded every alternative tariff at both endpoints and established the unique global optimum in Proposition 4.

C.4. Outcomes, exact bounds, and an open neighborhood

At τ_*, the before profile is C/C and the after profile is C/F, each with full participation almost surely. High-class return rises from g to $2g - 0.1$, and its conditional success falls from α to β_+. Low-class outcomes are unchanged. Inserting these values into (6)–(7) gives the changes in Table 2. The certificate encloses all five differences and establishes their strict signs.

The certificate uses exact rational bounds for exponentials. For $x \geq 0$, let $S_N(x)$ be the Taylor sum through order N. The next-term ratio bounds the tail whenever $x < N + 2$:

$$S_N(x) \leq e^x \leq S_N(x) + \frac{x^{N+1}/(N+1)!}{1 - x/(N+2)}. \tag{C12}$$

Reciprocation bounds negative exponentials. Monotonicity of se^s and the rational exponential bounds verify the Lambert-root brackets. These exact inequalities establish the strict outcome signs.

The distant-tariff bounds, local one-sided slopes, workflow preferences, and outcome signs are strict. They persist for sufficiently small changes in continuous primitives. The full-demand kink also

moves continuously with those primitives. These conditions establish a nonempty relative open neighborhood with the same kink optimum and adverse directions. The claim preserves the declared timing, available contracts, and support-family structure. It makes no positive-gap claim about tariffs arbitrarily close to the optimum and does not extend the rational certificate to the complete numerical grid.

Appendix D. Market-share support and ties

Subtract aggregate profit at an alternative tariff from profit at τ_0. With the notation of (17), the difference is $\lambda d(\tau) - (1-\lambda)b(\tau)$. Under uniqueness of the low-class maximizer, $d(\tau) > 0$ at every alternative. Dividing by $(1-\lambda)d(\tau)$ proves (18) for $0 < \lambda < 1$.

Whenever the condition holds, the complete maximizing set is

$$\{\tau_0\} \cup \left\{ \tau \neq \tau_0 : \frac{b(\tau)}{d(\tau)} = \frac{\lambda}{1-\lambda} \right\}. \tag{D1}$$

Strict inequality against the supremum gives uniqueness. Equality may still give uniqueness if no other tariff attains the supremum. If $\Theta < 0$, every interior share satisfies the condition; equivalently use $\max\{0, \Theta\}$ for a nonnegative share threshold. If $\Theta = +\infty$, no share below one supports τ_0. At share one, the low-market maximizer is uniquely optimal; at share zero, it must separately maximize high-market profit.

If the low-class maximizer is nonunique, the ratio condition applies only where $d > 0$. At alternatives with $d = 0$, it is necessary and sufficient that $b \leq 0$. A positive numerator at a zero denominator excludes τ_0 for every interior share. Zero numerator and denominator give a tie at every share. A negative numerator at zero denominator never blocks the candidate for an interior share.

Compactness, continuity, and a unique low-market optimum bound the ratio away from any neighborhood of τ_0. Its possible divergence is local. Suppose on each side of the candidate,

$$d(\tau_0 + x) = a_\pm |x|^{p_\pm} + o(|x|^{p_\pm}), \quad b(\tau_0 + x) = b_\pm |x|^{q_\pm} + o(|x|^{q_\pm}), \quad a_\pm > 0. \tag{D2}$$

On a side with $b_\pm > 0$, the ratio is bounded when $q_\pm \geq p_\pm$ and diverges positively when $q_\pm < p_\pm$. A negative leading numerator can diverge negatively without invalidating an upper bound. A strict kink with nonzero one-sided slopes gives a first-order lower bound for d. Locally Lipschitz H gives a first-order upper bound for $|b|$, so the ratio is bounded.

At a smooth interior low-market maximum, $L'(\tau_0) = 0$ and $L''(\tau_0) < 0$. If $H'(\tau_0) \neq 0$, one side has a positive first-order numerator and a second-order denominator; hence $\Theta = +\infty$. If $H'(\tau_0) = 0$ and H is twice differentiable, the local ratio tends to $H''(\tau_0)/[-L''(\tau_0)]$. Higher-order flat maxima require the corresponding order comparison. The threshold uses the complete optimized class-profit functions, including both discovery costs, and has no generic monotonicity in either cost.

Appendix E. Execution-fee optimization and contract choice

E.1. Proof of Proposition 5

The transformation in (19) maps nonnegative (f, τ) bijectively to the closed bill cone. Fix a mixed workflow profile. A component F_i is unimodal when it is nondecreasing up to its interval of global maximizers and nonincreasing thereafter. Choose a finite $M > 0$ at least as large as one global maximizer of each component. If $p_1 > M$, reduce both bills proportionally until $p_1 = M$. Both prices remain above those component maximizers, so unimodality prevents a profit decline. The search can therefore be restricted to the compact intersection of the cone and $[0, M] \times [0, 2M]$, where continuity gives a global maximum.

Start at such a constrained maximum. If it is on a cone boundary, it is already one of the stated candidates. Otherwise move p_1 toward its component-maximizing interval, holding p_2 fixed, until reaching that interval or a boundary. Unimodality prevents a decline in the component objective. Since the starting point is globally optimal, the total objective remains at its maximum.

If the first move reaches a boundary, the proof is complete. Otherwise p_1 is individually optimal. Move p_2 toward its own maximizing interval in the same way. This reaches either a boundary or a pair of independent component maximizers. Hence the candidate sets contain a representative global optimum. With strict monotonicity on each side of active maximizing intervals, every strictly interior active optimum has both components individually optimal.

Weak unimodality can create additional coordinate ties. One component may have an inactive plateau below a distant higher peak, while the other component's peak lies in the price region compatible with that plateau. Interior pairs can then tie a boundary representative without maximizing the plateau component independently. The representative result remains valid, but these pairs need not be uniquely identified. The calculation records zero-discovery and workflow ties rather than interpreting unused prices as unique active choices.

For C/C, only p_1 enters profit, and a boundary can supply any feasible unused p_2. For F/F, only p_2 matters, with $p_1 \in [p_2/2, p_2]$. A single active class can likewise achieve its best bill with an unused coordinate on a boundary. OFF/OFF produces zero. The mixed profiles C/F and F/C use the preceding argument. These cases exhaust the nine profiles. Maximization over a finite profile set commutes with maximization over prices, which establishes the global value of the complete problem.

At an optimal fee/rate pair, a strictly profitable continuation workflow change would be another profile already included in the maximization. It therefore cannot improve the global value. All tied profiles remain available. The equality $p_1 = p_2$ describes a boundary of execution bills; interpreting it as a separately certified genuine-goal contract requires the corresponding observability assumptions in addition to the algebra.

E.2. Complete certification-cost comparisons

Let $K_j = \Pi^j_{AS} - \Pi^j_B$ for quality endpoint j. In the examples, Π_{AS} is unchanged because the coarse technology serves both classes at both endpoints. Net supplier profit is $\max\{\Pi^j_B, \Pi^j_{AS} - K_T\}$. Welfare subtracts K_T exactly when certification is purchased. User surplus and correct-goal counts have no direct certificate-cost deduction.

For Example 2, $K_{\text{after}} = K_- < K_+ = K_{\text{before}}$. Below both thresholds, the same class-pricing allocation is selected twice. Above both, the two fee outcomes are compared. Between them, the change is from class pricing before improvement to fee pricing afterwards. Subtracting the contract outcomes gives (21) and the middle-region user and correct-goal differences. At K_-, the after contract can be either alternative; at K_+, the before contract can be either. Their outcome

differences are exactly the adjacent-region endpoint differences. The distributed certification-region table includes each such equality choice for both examples.

Appendix F. Discovery proofs and price feedback

F.1. Optimized responses and Proposition 6

Dividing the objective in (6) by k shows that the optimizer depends on $t = R/k$. Differentiating the first-order condition gives

$$\frac{ds}{dt} = \frac{r'(s)}{\mathcal{C}''(s) - tr''(s)} > 0. \tag{F1}$$

Multiplication by $r'(s)$ gives $\rho'(t)$, and multiplication by $t/\rho(t)$ proves the elasticity expression in (22). Strict concavity of the objective at the optimum makes the denominator positive. The reduced-form assumptions in (23), including decreasing elasticity, are additional conditions.

Differentiate (24) with respect to $\log k$. The chain rule gives $\varepsilon(R_C/k) - \varepsilon(R_F/k) > 0$. As k tends to zero, both discovery probabilities approach one. As k tends to infinity, their asymptotic power forms give the limit A_R^η. Continuity and strict monotonicity imply that the range is exactly the open interval between those limits.

For positive x_C, x_F, compare $X_F/X_C = H(k)/d$. If $d \leq 1$, that ratio exceeds one for every finite cost. If $d \geq A_R^\eta$, it is below one for every finite cost. Between the two limits, the intermediate value theorem and strict monotonicity give one equality and the stated loss region. Zero conditional outcomes are compared directly without division. This proves Proposition 6.

F.2. Proof of Proposition 7

The envelope theorem yields $\Psi_R(R, k) = \rho(R/k)$. Integrate this derivative from R_C to R_F and add the user-surplus difference to obtain (25). For any fixed $R > R_C$, the same response-ratio argument makes $\rho(R/k)/\rho(R_C/k)$ strictly increasing in k. The integral is therefore strictly increasing, even if $u_F = 0$.

The finite-cost ratios satisfy

$$1 \leq \frac{\rho(R/k)}{\rho(R_C/k)} \leq \left(\frac{R}{R_C}\right)^{\eta}, \qquad R \in [R_C, R_F]. \tag{F2}$$

Strict inequalities hold for interior $R > R_C$ at finite k. The upper bound is integrable on the compact interval. Dominated convergence makes the increasing component in (25) tend to L_W at zero cost and to M_W at infinite cost. Since $R_F > R_C$ and $\eta > 0$, $M_W > L_W$. Comparing u_C with these limits proves every case in Proposition 7, including the endpoint cases.

The supplier-profit difference is strictly positive because $\Psi_R > 0$ on $[R_C, R_F]$. Thus $W_F - W_C = (U_F - U_C) + (\Psi_F - \Psi_C) < 0$ implies $U_F - U_C < 0$. If both cutoffs are finite, welfare is positive at the user-surplus equality. Each normalized difference crosses once from negative to positive, so $k_W < k_U$. A correct-goal cutoff uses a different conditional ratio and has no general ordering with these two cutoffs.

For the compensation statement, uniqueness and $s_F > s_C$ imply $\Pi_F(s_F) > \Pi_F(s_C)$. Increasing discovery and $u_F \geq 0$ imply $[r(s_F) - r(s_C)]u_F \geq 0$. Adding the two differences proves (27). This argument holds the new conditional offer fixed and compares two discovery efforts at that offer.

F.3. The primitive families

For exponential discovery with power cost, the first-order condition gives $t = s^{p-1}e^s$. Its positive solution is

$$s = (p-1)W_0\left(\frac{t^{1/(p-1)}}{p-1}\right). \tag{F3}$$

The elasticity in (28) has logarithmic derivative with respect to s

$$\frac{p-1}{s(s+p-1)} - \frac{1}{1-e^{-s}} < \frac{1}{s} - \frac{1}{1-e^{-s}} < 0. \tag{F4}$$

Since s increases with t, elasticity decreases with t. Near zero, $s \sim t^{1/(p-1)}$ and $\rho \sim s$; at infinity, s diverges and discovery tends to one. Thus (23) holds with $a_0 = 1$ and $\eta = 1/(p-1)$.

For hyperbolic discovery, the first-order condition is $t = s^{p-1}(1+s)^2$, strictly increasing in s. Direct substitution in (22) yields (29), which strictly decreases with s. The same small-s asymptotic gives the same exponent, and discovery tends to one as effort diverges. Both families therefore satisfy the complete theorem assumptions for every $p > 1$.

For the counterexample cost in (30), $\mathcal{C}'(s) = s + 1 - e^{-10s}$ and $\mathcal{C}''(s) = 1 + 10e^{-10s}$. Exponential discovery is strictly concave, so the objective has a unique interior maximum. Its first-order condition and elasticity have expansions

$$t = e^s(s + 1 - e^{-10s}) = 11s - 39s^2 + O(s^3), \tag{F5}$$

$$\varepsilon(t(s)) = \frac{s + 1 - e^{-10s}}{(e^s - 1)(s + 2 + 9e^{-10s})} = 1 + \frac{67}{22}s + O(s^2). \tag{F6}$$

Inverting the first expansion and substituting into discovery gives $\rho(t) = t/11 + 67t^2/2662 + O(t^3)$. The ratio at fixed positive returns then yields (31). Smoothness of these analytic expansions also establishes eventual decline of H toward A_R, so the counterexample contradicts monotonicity while retaining strictly convex cost and a unique discovery optimum.

F.4. Endogenous tariffs

On a smooth active profile,

$$\Pi_\tau = \sum_i \lambda_i \rho(R_i/k_i) R_i', \quad \Pi_{\tau\tau} = \sum_i \lambda_i \left[\frac{\rho'(R_i/k_i)}{k_i} (R_i')^2 + \rho(R_i/k_i) R_i'' \right]. \tag{F7}$$

At a fixed tariff, $\partial \rho(R_z/k_z)/\partial k_z = -\rho(R_z/k_z)\varepsilon(R_z/k_z)/k_z$. Differentiating $\Pi_\tau(\tau^*, k_z) = 0$ and dividing by the negative second derivative proves (32). The numerator's sign is the conditional-return slope, giving the stated opposite sign for the tariff response. At a nonsmooth or nonunique optimum, this local argument does not apply.

If both compared returns depend on a common cost, apply the chain rule to $\log \rho(R_m(k)/k)$ for each offer. Subtracting the two derivatives gives (33). For a conditional outcome $x_m(k)$, the derivative

of $\log x_F - \log x_C$ must also be added to the outcome-ratio derivative. These terms specify why changing endogenous prices or conditional verification requires a full equilibrium calculation.

Appendix G. Computational methods and complete comparisons

G.1. Finite calculation domains

The complete uniform-value grid crosses six axes:

$$\begin{aligned} \lambda \in \{0.5, 0.65, 0.8, 0.95\}, \quad k_L, k_H \in \{0.001, 0.01, 0.1\}, \\ \kappa, \nu \in \{0, 0.05, 0.1\}, \qquad\qquad \omega \in \{0.01, 0.02\}. \end{aligned} \tag{G0}$$

Their Cartesian product contains 648 combinations. Each is evaluated at fine-step errors 0.35 and 0.10, giving 1,296 environments. The shared review technology, value supports, and coarse error are in Table 1. The supplier solves common action pricing, common goal pricing, and class-specific pricing in every environment. The class-specific action and goal solutions coincide by Proposition 2 and are also directly compared in the computation.

The additional domains retain the same primitives except for their declared axis. The six value shapes generate 24 endpoint environments across the two examples. The full quality path generates 282 environments. The real-cost comparison generates eight. The market-share path uses 100 shares from 0.5 to 0.995, spaced by 0.005, at both quality endpoints. An additional 72 environments cross four shares with both discovery-cost scales at three values and the two endpoints. The execution-fee comparison covers both examples and both endpoints. General discovery uses 201 logarithmically spaced costs from 0.0001 to 10 for each of six primitive families, plus the separate 201-point convex-cost counterexample.

Outcome counts use a numerical equality tolerance of 10^{-9}. They summarize the specified grid. The result files include workflow ties and price covers alongside the reported optimum.

G.2. Global tariff computation

The uniform solver uses the piecewise formulas in Appendix A. On each demand segment, conditional return is linear or quadratic in the tariff. Discovery value is increasing in conditional return, which supplies interval profit bounds. The algorithm evaluates feasible candidates and subdivides intervals until the best upper profit bound is within 10^{-9} of the best feasible profit.

The full solver begins with the verification solution in Appendix A. Since h_n is increasing, a scalar root identifies the acceptance cutoff. Integer-parameter Beta densities are polynomials in normalized value $(v-\ell)/(u-\ell)$. On zero-review and full-review segments, the return bound is obtained from that polynomial's endpoints and stationary points. This normalized representation avoids cancellation from unnecessarily large coefficients in unscaled values.

Write $S_n(v) = P_n(b_n(v))$ for success after optimized review, and let f_z denote the value density. On a partial-demand segment where the marginal customer chooses interior review, the derivative identities are

$$Q_p = -\frac{f_z(v_p)}{S_n(v_p)}, \qquad Q_{pp} = -\frac{f_z'(v_p)}{S_n(v_p)^2} + \frac{f_z(v_p)S_n'(v_p)}{S_n(v_p)^3}. \tag{G1}$$

To make the bound explicit, write $p = d_p\tau$, where $d_p = n$ under action pricing and $d_p = 1$ under total-price billing, and let $m = n\kappa$. Suppose $f_z \le f_{\max}$, $|f_z'| \le f_{\max}'$, $S_n \ge S_{\min}$, and $0 \le S_n' \le V_{\max}$ on the segment. A bound on the absolute tariff second derivative of conditional return is

$$M_R = d_p^2\left[\frac{2f_{\max}}{S_{\min}} + \max_{\tau\in[a_t,b_t]}|d_p\tau - m|\left(\frac{f_{\max}'}{S_{\min}^2} + \frac{f_{\max}V_{\max}}{S_{\min}^3}\right)\right]. \tag{G2}$$

The implementation uses $S_{\min} = (1-\chi)(1-e_n)^n$. Differentiating the interior-review solution gives the valid bounds $V_{\max} = \omega/(a\ell^2)$ for one action and $V_{\max} = 2\omega(1-e_n)/[a\ell^2(1-2e_n)]$ for two. The numerical examples use $a = 1$. Polynomial density extrema supply the two density bounds. The conditional-return upper bound is then

$$\max\{R(a_t), R(b_t)\} + M_R(b_t - a_t)^2/8. \tag{G3}$$

Equation (G3) follows from the endpoint interpolation error bound $M_R(\tau - a_t)(b_t - \tau)/2 \leq M_R(b_t - a_t)^2/8$. The solver transforms each class's return bound through its increasing discovery value and sums the weighted bounds. The full-model target profit gap is 10^{-8}, with at most 200,000 subdivisions. Candidate local optimizations improve feasible lower bounds; they do not replace the interval comparison. Breakpoints include review transitions and support boundaries. Outcome integration is split at the same value transitions, with absolute and relative tolerances 2×10^{-10}.

Class-specific pricing is solved one class at a time. The execution-fee problem uses Proposition 5 and all nine workflow profiles. Common-price boundaries still undergo global scalar optimization. Saved output includes the tariff, workflow choices, conditional quantities, discovery, real-cost components, reported-price workflow ties, and the remaining outer cover for optimal prices.

The grid calculation uses floating-point profit bounds; their stopping gap measures objective resolution. Welfare and price differences at tied optima need separate comparison. Appendix C's rational enclosures establish the strict inequalities for Example 2.

Array operations use NumPy (Harris et al., 2020), and integration and optimization use SciPy (Virtanen et al., 2020). Symbolic identities use SymPy (Meurer et al., 2017), and the figures use Matplotlib (Hunter, 2007).

G.3. Numerical comparisons

The following tables report the full named comparisons. Values are rounded for display; the replication files preserve full precision. A negative change always means the after-improvement equilibrium minus the corresponding before-improvement equilibrium. Contract-specific profits and welfare are gross of certificate costs unless the caption explicitly states otherwise.

Table G1. Complete contract outcomes in the two examples.

Ex.	Quality	Contract	Profile	Π	U	W	X
1	before	A	C/F	0.200783	0.046208	0.246992	0.729346
1	before	G	C/C	0.185411	0.137695	0.323106	0.786827
1	before	AS	C/C	0.292096	0.035662	0.327758	0.797017
1	before	B	C/F	0.213152	0.034063	0.247216	0.733074
1	after	A	C/F	0.219571	0.079470	0.299041	0.784101
1	after	G	C/C	0.185411	0.137695	0.323106	0.786827
1	after	AS	C/C	0.292096	0.035662	0.327758	0.797017
1	after	B	C/F	0.219571	0.079470	0.299041	0.784101
2	before	A	C/C	0.183902	0.149675	0.333577	0.855287
2	before	G	C/C	0.183902	0.149675	0.333577	0.855287
2	before	AS	C/C	0.296144	0.038546	0.334691	0.857630
2	before	B	C/F	0.193929	0.036948	0.230877	0.793685
2	after	A	C/F	0.202671	0.082825	0.285496	0.847206
2	after	G	C/C	0.183902	0.149675	0.333577	0.855287
2	after	AS	C/C	0.296144	0.038546	0.334691	0.857630
2	after	B	C/F	0.202671	0.082825	0.285496	0.847206

Notes: Profile lists low / high workflows. D duplicates AS by Proposition 2. B includes the execution fee. Profits and welfare exclude certificate costs. The supplied table files also report rates, fees, and billable actions.

Table G2. Private and social values of type certification.

Ex.	Quality	Private, A	Social, A	Private, B	Social, B
1	before	0.091313	0.080767	0.078944	0.080543
1	after	0.072525	0.028717	0.072525	0.028717
2	before	0.112242	0.001114	0.102215	0.103814
2	after	0.093474	0.049195	0.093474	0.049195

Notes: A and B denote the available outside contract. Private values are $\Pi_{AS} - \Pi_j$; social values are $W_{AS} - W_j$, before deducting K_T. These differences need not coincide.

Table G3. All demand-shape comparisons under common action pricing.

Ex.	Beta shape	τ_b	τ_a	ΔU	ΔW	ΔX
1	(1, 1)	0.268264	0.285463	+0.033262	+0.052049	+0.054755
1	(2, 2)	0.268709	0.294073	+0.024473	+0.047653	+0.038498
1	(2, 1)	0.269550	0.307075	+0.012021	+0.042431	+0.018608
1	(1, 2)	0.268264	0.285463	+0.032774	+0.051562	+0.054755
1	(3, 1)	0.317062	0.320044	-0.068223	-0.027633	-0.012572
1	(1, 3)	0.268264	0.285463	+0.032530	+0.051318	+0.054755
2	(1, 1)	0.295463	0.295463	-0.066850	-0.048081	-0.008081
2	(2, 2)	0.303898	0.305706	-0.070650	-0.050031	-0.016230
2	(2, 1)	0.316445	0.320098	-0.075298	-0.052003	-0.023116
2	(1, 2)	0.295463	0.295463	-0.066783	-0.048014	-0.008081
2	(3, 1)	0.329389	0.332362	-0.077209	-0.051409	-0.019683
2	(1, 3)	0.295463	0.295463	-0.066749	-0.047980	-0.008081

Notes: Subscripts b and a denote before and after improvement. Each Beta distribution is scaled to the class-specific support in Table 1. Every row reoptimizes the common rate, workflow choices, and discovery.

Table G4. Zero fine-step error relative to the initial technology.

Ex.	Outcome	At error 0.35	At error 0	Change
1	User surplus	0.046208	0.106057	+0.059849
1	Welfare	0.246992	0.325628	+0.078636
1	Correct goals	0.729346	0.807121	+0.077775
2	User surplus	0.149675	0.109269	-0.040407
2	Welfare	0.333577	0.311939	-0.021638
2	Correct goals	0.855287	0.871467	+0.016180

Notes: Common action pricing with fully optimized verification. Zero fine-step error does not remove the common failure probability.

Table G5. Direction counts across all 648 quality comparisons.

Outcome	Decrease	Increase	Unchanged
Supplier profit	0	385	263
User surplus	177	208	263
Welfare	171	214	263
Correct goals	113	272	263
Billable actions	0	385	263

Notes: Counts concern common action pricing and reported optimal representatives. Differences within 10^{-9} are classified as unchanged. Common goal and class-specific pricing have 648 unchanged comparisons for every displayed outcome. The domain is given in equation (G0).

Table G6. All outcome changes at type-certification cost equalities.

Ex.	K_T	Contract pair	ΔU	ΔW	ΔX
1	0.072525	AS to AS	+0.000000	+0.000000	+0.000000
1	0.072525	AS to B	+0.043808	+0.043808	-0.012916
1	0.078944	AS to B	+0.043808	+0.050227	-0.012916
1	0.078944	B to B	+0.045406	+0.051825	+0.051028
2	0.093474	AS to AS	+0.000000	+0.000000	+0.000000
2	0.093474	AS to B	+0.044278	+0.044278	-0.010424
2	0.102215	AS to B	+0.044278	+0.053020	-0.010424
2	0.102215	B to B	+0.045877	+0.054619	+0.053521

Notes: AS denotes class-specific action pricing and B the execution-fee contract. Welfare deducts K_T whenever certification is purchased. Each row is an optimal before/after contract pair at the specified equality; both adjacent alternatives are retained.

Generative AI Use Statement

ChatGPT (OpenAI) was used solely for grammar checking, language polishing, and LaTeX typesetting during the preparation of this manuscript. The final manuscript was reviewed and edited by the author, who takes full responsibility for its content.

References

Agrawal, A., Gans, J., & Goldfarb, A. (2018). *Prediction, judgment and complexity: A theory of decision making and artificial intelligence* (NBER Working Paper No. 24243). National Bureau of Economic Research. https://doi.org/10.3386/w24243

Anderson, S. P., & de Palma, A. (2009). Information congestion. *The RAND Journal of Economics*, *40*(4), 688–709. https://doi.org/10.1111/j.1756-2171.2009.00085.x

Baird, A., & Maruping, L. M. (2021). The next generation of research on IS use: A theoretical framework of delegation to and from agentic IS artifacts. *MIS Quarterly*, *45*(1), 315–341. https://doi.org/10.25300/misq/2021/15882

Berman, O., & Ashrafi, N. (1993). Optimization models for reliability of modular software systems. *IEEE Transactions on Software Engineering*, *19*(11), 1119–1123. https://doi.org/10.1109/32.256858

Bhargava, H. K., & Choudhary, V. (2008). Research note: When is versioning optimal for information goods? *Management Science*, *54*(5), 1029–1035. https://doi.org/10.1287/mnsc.1070.0773

Bryan, K., & Gans, J. (2026). *Training AI for when humans will use it* (NBER Working Paper No. 35490). National Bureau of Economic Research. https://doi.org/10.3386/w35490

Cao, Y., Chen, Y., Ding, Y., & Zhang, T. (2026). Search and competition in expert markets. *The RAND Journal of Economics*, *57*(1), 158–175. https://doi.org/10.1111/1756-2171.70049

Corless, R. M., Gonnet, G. H., Hare, D. E. G., Jeffrey, D. J., & Knuth, D. E. (1996). On the Lambert W function. *Advances in Computational Mathematics*, *5*(1), 329–359. https://doi.org/10.1007/bf02124750

Deneckere, R. J., & McAfee, R. P. (1996). Damaged goods. *Journal of Economics & Management Strategy*, *5*(2), 149–174. https://doi.org/10.1111/j.1430-9134.1996.00149.x

Dulleck, U., & Kerschbamer, R. (2006). On doctors, mechanics, and computer specialists: The economics of credence goods. *Journal of Economic Literature*, *44*(1), 5–42. https://doi.org/10.1257/002205106776162717

Dulleck, U., & Kerschbamer, R. (2009). Experts vs. discounters: Consumer free-riding and experts withholding advice in markets for credence goods. *International Journal of Industrial Organization*, *27*(1), 15–23. https://doi.org/10.1016/j.ijindorg.2008.02.005

Dulleck, U., Kerschbamer, R., & Sutter, M. (2011). The economics of credence goods: An experiment on the role of liability, verifiability, reputation, and competition. *American Economic Review*, *101*(2), 526–555. https://doi.org/10.1257/aer.101.2.526

Emons, W. (1997). Credence goods and fraudulent experts. *The RAND Journal of Economics*, *28*(1), 107–119. https://doi.org/10.2307/2555942

Erlei, A. (2025). *From digital distrust to codified honesty: Experimental evidence on generative AI in credence goods markets* [Preprint]. arXiv. https://doi.org/10.48550/arXiv.2509.06069

Harris, C. R., Millman, K. J., van der Walt, S. J., Gommers, R., Virtanen, P., Cournapeau, D., Wieser, E., Taylor, J., Berg, S., Smith, N. J., Kern, R., Picus, M., Hoyer, S., van Kerkwijk, M. H., Brett, M., Haldane, A., del Río, J. F., Wiebe, M., Peterson, P., ... Oliphant, T. E. (2020). Array programming with NumPy. *Nature*, *585*(7825), 357–362. https://doi.org/10.1038/s41586-020-2649-2

Holmstrom, B., & Milgrom, P. (1991). Multitask principal-agent analyses: Incentive contracts, asset ownership, and job design. *The Journal of Law, Economics, and Organization*, *7*(Special Issue), 24–52. https://doi.org/10.1093/jleo/7.special_issue.24

Hu, X., & Lei, H. (2023). *Credibility in credence goods markets* [Preprint]. arXiv. https://doi.org/10.48550/arXiv.2310.09544

Huang, J., Chen, X., Mishra, S., Zheng, H. S., Yu, A. W., Song, X., & Zhou, D. (2024). Large language models cannot self-correct reasoning yet. *International Conference on Learning Representations*. https://openreview.net/forum?id=IkmD3fKBPQ

Huang, L., Xiao, W., & Vishnoi, N. K. (2026). *Delegation and verification under AI* [ICML 2026 accepted paper; arXiv version 2]. arXiv. https://doi.org/10.48550/arXiv.2603.02961

Hunter, J. D. (2007). Matplotlib: A 2D graphics environment. *Computing in Science & Engineering*, *9*(3), 90–95. https://doi.org/10.1109/mcse.2007.55

Jeong, S., Baek, J., Kang, M., & Hwang, S. J. (2026). *TIDE: Proactive multi-problem discovery via template-guided iteration* [Preprint]. arXiv. https://doi.org/10.48550/arXiv.2606.04743

Jimenez, C. E., Yang, J., Wettig, A., Yao, S., Pei, K., Press, O., & Narasimhan, K. (2024). SWE-bench: Can language models resolve real-world GitHub issues? *International Conference on Learning Representations*. https://openreview.net/forum?id=VTF8yNQM66

Khot, T., Trivedi, H., Finlayson, M., Fu, Y., Richardson, K., Clark, P., & Sabharwal, A. (2023). Decomposed prompting: A modular approach for solving complex tasks. *International Conference on Learning Representations*. https://openreview.net/forum?id=_nGgzQjzaRy

Kim, E. M., Garg, A., Peng, K., & Garg, N. (2025). Correlated errors in large language models. *Proceedings of the 42nd International Conference on Machine Learning*, *267*, 30038–30066. https://proceedings.mlr.press/v267/kim25e.html

Kleinberg, J., & Raghavan, M. (2021). Algorithmic monoculture and social welfare. *Proceedings of the National Academy of Sciences*, *118*(22), Article e2018340118. https://doi.org/10.1073/pnas.2018340118

Knight, J. C., & Leveson, N. G. (1986). An experimental evaluation of the assumption of independence in multiversion programming. *IEEE Transactions on Software Engineering*, *SE-12*(1), 96–109. https://doi.org/10.1109/tse.1986.6312924

Lipnowski, E., Mathevet, L., & Wei, D. (2020). Attention management. *American Economic Review: Insights*, *2*(1), 17–32. https://doi.org/10.1257/aeri.20190165

Littlewood, B., & Miller, D. R. (1989). Conceptual modeling of coincident failures in multiversion software. *IEEE Transactions on Software Engineering*, *15*(12), 1596–1614. https://doi.org/10.1109/32.58771

McAfee, R. P. (2007). Pricing damaged goods. *Economics*, *1*(1), Article 2007-1. https://doi.org/10.5018/economics-ejournal.ja.2007-1

Meurer, A., Smith, C. P., Paprocki, M., Čertík, O., Kirpichev, S. B., Rocklin, M., Kumar, A., Ivanov, S., Moore, J. K., Singh, S., Rathnayake, T., Vig, S., Granger, B. E., Muller, R. P., Bonazzi, F., Gupta, H., Vats, S., Johansson, F., Pedregosa, F., … Scopatz, A. (2017). SymPy: Symbolic computing in Python. *PeerJ Computer Science*, *3*, Article e103. https://doi.org/10.7717/peerj-cs.103

Microsoft. (n.d.). *Billing rates and management*. Retrieved September 5, 2026, from https://learn.microsoft.com/en-us/microsoft-copilot-studio/requirements-messages-management

Mozannar, H., & Sontag, D. (2020). Consistent estimators for learning to defer to an expert. *Proceedings of the 37th International Conference on Machine Learning*, *119*, 7076–7087. https://proceedings.mlr.press/v119/mozannar20b.html

Mussa, M., & Rosen, S. (1978). Monopoly and product quality. *Journal of Economic Theory*, *18*(2), 301–317. https://doi.org/10.1016/0022-0531(78)90085-6

Pesendorfer, W., & Wolinsky, A. (2003). Second opinions and price competition: Inefficiency in the market for expert advice. *Review of Economic Studies*, *70*(2), 417–437. https://doi.org/10.1111/1467-937x.00250

Salesforce. (n.d.). *Agentforce pricing*. Retrieved September 5, 2026, from https://www.salesforce.com/agentforce/pricing/

Schick, T., Dwivedi-Yu, J., Dessì, R., Raileanu, R., Lomeli, M., Zettlemoyer, L., Cancedda, N., & Scialom, T. (2023). Toolformer: Language models can teach themselves to use tools. *Advances in Neural Information Processing Systems*, *36*. https://arxiv.org/abs/2302.04761

Sims, C. A. (2003). Implications of rational inattention. *Journal of Monetary Economics*, *50*(3), 665–690. https://doi.org/10.1016/s0304-3932(03)00029-1

Sundararajan, A. (2004). Nonlinear pricing of information goods. *Management Science*, *50*(12), 1660–1673. https://doi.org/10.1287/mnsc.1040.0291

Virtanen, P., Gommers, R., Oliphant, T. E., Haberland, M., Reddy, T., Cournapeau, D., Burovski, E., Peterson, P., Weckesser, W., Bright, J., van der Walt, S. J., Brett, M., Wilson, J., Millman, K. J., Mayorov, N., Nelson, A. R. J., Jones, E., Kern, R., Larson, E., ... Vázquez-Baeza, Y. (2020). SciPy 1.0: Fundamental algorithms for scientific computing in Python. *Nature Methods*, *17*(3), 261–272. https://doi.org/10.1038/s41592-019-0686-2

Weitzman, M. L. (1979). Optimal search for the best alternative. *Econometrica*, *47*(3), 641–654. https://doi.org/10.2307/1910412

White, A., Jain, K., Ichihashi, S., & Kim, B. (2024). Double marginalization and misplacement in online advertising. *The Journal of Industrial Economics*, *72*(2), 940–964. https://doi.org/10.1111/joie.12376

Yang, Y.-T., & Zhu, Q. (2025). PACT: A contract-theoretic framework for pricing agentic AI services powered by large language models. *GLOBECOM 2025 - 2025 IEEE Global Communications Conference*, 5677–5682. https://doi.org/10.1109/globecom59602.2025.11431824

Yao, S., Zhao, J., Yu, D., Du, N., Shafran, I., Narasimhan, K., & Cao, Y. (2023). ReAct: Synergizing reasoning and acting in language models. *International Conference on Learning Representations*. https://arxiv.org/abs/2210.03629

Ye, L., & Zhang, C. (2017). Monopolistic nonlinear pricing with consumer entry. *Theoretical Economics*, *12*(1), 141–173. https://doi.org/10.3982/te1944

Zhou, D., Schärli, N., Hou, L., Wei, J., Scales, N., Wang, X., Schuurmans, D., Cui, C., Bousquet, O., Le, Q., & Chi, E. (2023). Least-to-most prompting enables complex reasoning in large language models. *International Conference on Learning Representations*. https://openreview.net/forum?id=WZH7099tgfM

Zhou, J., Roy, A., Gupta, S., Weitekamp, D., & MacLellan, C. J. (2026). When should users check? Modeling confirmation frequency in multi-step agentic AI tasks. *Proceedings of the 2026 CHI Conference on Human Factors in Computing Systems*, 1–20. https://doi.org/10.1145/3772318.3790655

Zhou, S., Xu, F. F., Zhu, H., Zhou, X., Lo, R., Sridhar, A., Cheng, X., Ou, T., Bisk, Y., Fried, D., Alon, U., & Neubig, G. (2024). WebArena: A realistic web environment for building autonomous agents. *International Conference on Learning Representations*. https://arxiv.org/abs/2307.13854